



Implement Static Routes

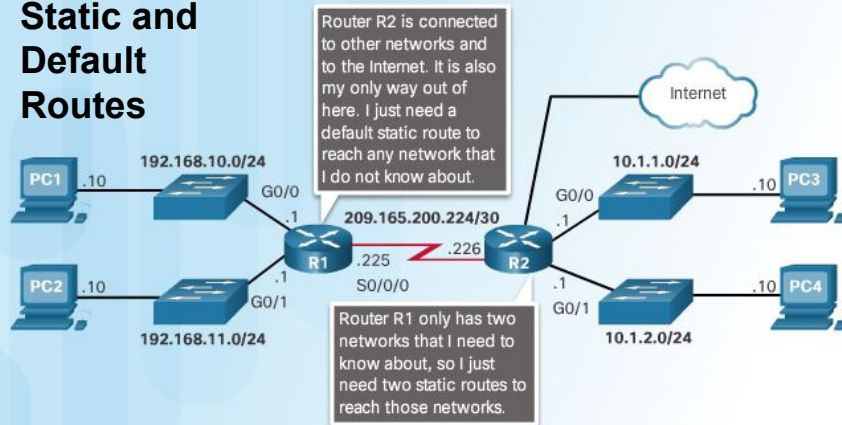
Static Routing: Reach Remote Networks



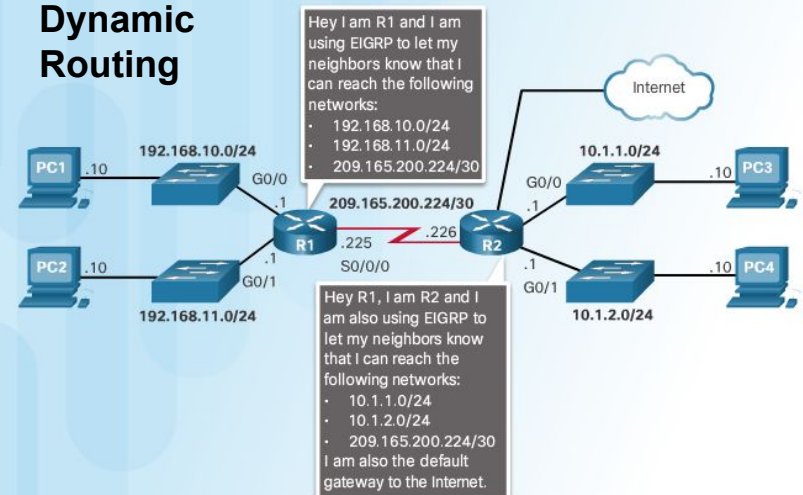
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- A router learns about remote networks in two ways:
 - Manually entered into the route table using static routes
 - Static routes are not automatically updated and must be reconfigured when topology changes
 - Dynamically (Automatically) learned using a routing protocol

Static and Default Routes



Dynamic Routing



Static Routing: Why Use Static Routing?



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Dynamic versus Static Routing

	Dynamic Routing	Static Routing
Configuration Complexity	Generally independent of the network size	Increases with network size
Topology Changes	Automatically adapts to topology changes	Administrator intervention required
Scaling	Suitable for simple and complex topologies	Suitable for simple topologies
Security	Less secure	More secure
Resource Usage	Uses CPU, memory, link bandwidth	No extra resources needed
Predictability	Route depends on the current topology	Route to destination is always the same

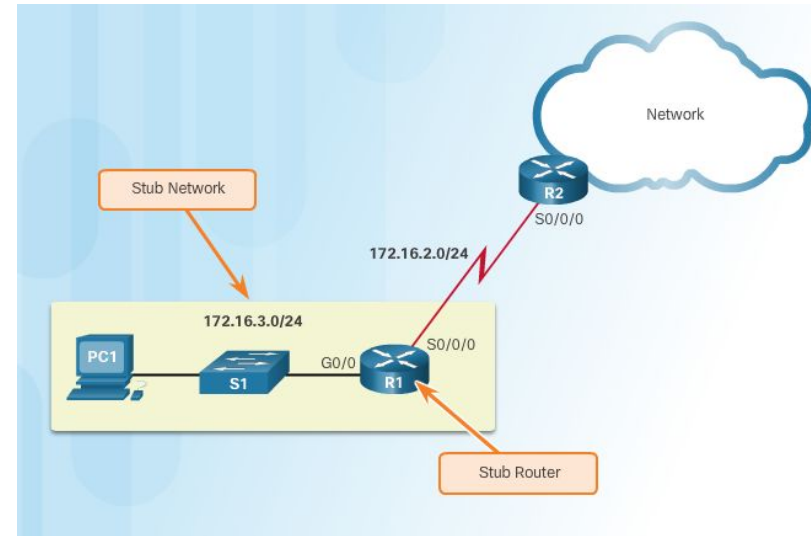
Static Routing: When to Use Static Routes



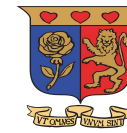
Three uses for static routes:

- Smaller networks that are not expected to grow
- Routing to and from stub networks
 - Stub network accessed by a single route and has one neighbor
 - 172.16.3.0 is a stub network
- A single default route to represent a path to any network not found in the routing table
 - Use default route on R1 to point to R2 for all other networks

Stub Networks and Stub Routers



Types of Static Routes: Static Route Applications



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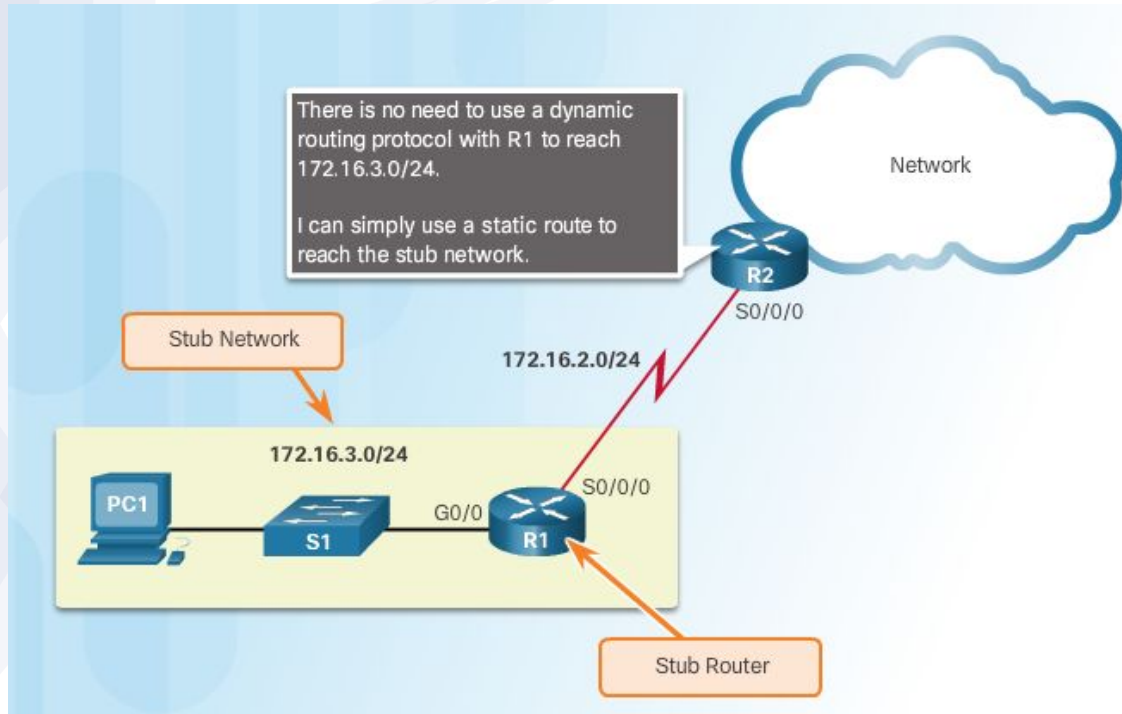
Use Static Routes To:

- Connect to a specific network
- Connect a stub router
- Summarize routing table entries which reduces size of routing advertisements
- Create a backup route in case a primary route link fails

Types of Static Routes: Standard Static Route



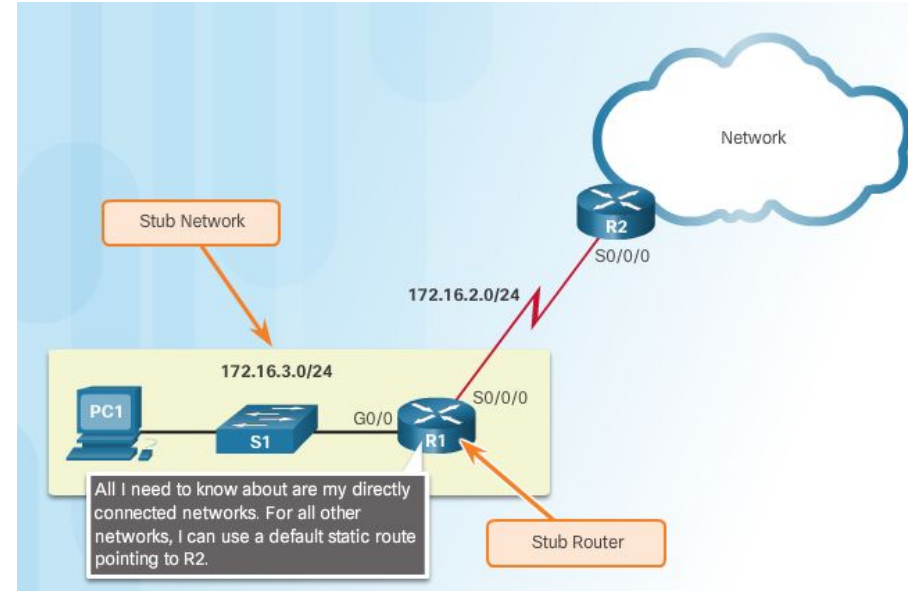
- R2 configured with a static route to reach the stub network 172.16.3.0/24



Types of Static Routes: Default Static Route



- Default route matches all packets and is used when a packet does not match a specific route in the routing table
- Can be dynamically learned or statically configured
- Default Static route uses 0.0.0.0/0 as the destination IPv4 address
- Creates a Gateway of Last Resort
- Common use is when connecting a company's edge router to the ISP network
- Router has only one router to which it is connected

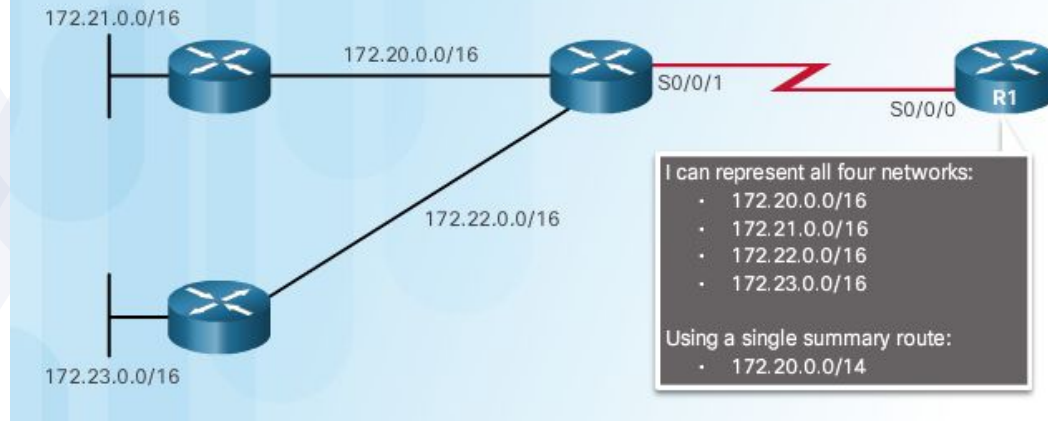


Types of Static Routes: Summary Static Route



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- Multiple static routes can be summarized into a single network address
 - Destination networks must be contiguous
 - Multiple static routes must use the same exit interface or next hop
 - In figure, four networks is summarized into one summary static route

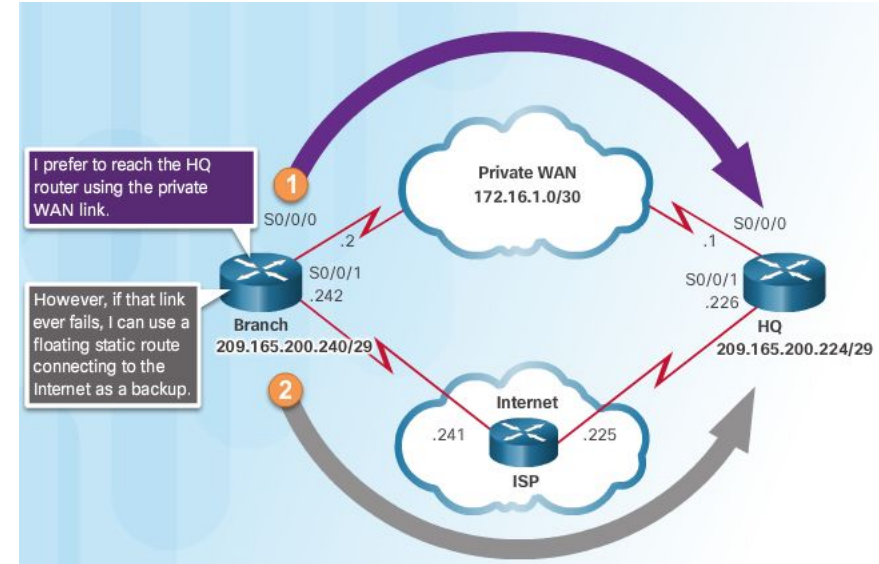


Types of Static Routes: Floating Static Route



- Static routes that are used to provide a backup path
- Used when primary route is not available
- Configured with a higher administrative distance (trustworthiness) than the primary route
- Example: EIGRP administrative distance equals 90. A floating static route with an AD of 91 or higher would serve as backup route and will be used if EIGRP route goes down.

Floating Static Route





Configure Static and Default Routes

Configure IPv4 Static Routes: ip route Command



ip route Command Syntax

```
Router(config)# ip route network-address subnet-mask {ip-address | exit-intf}
```

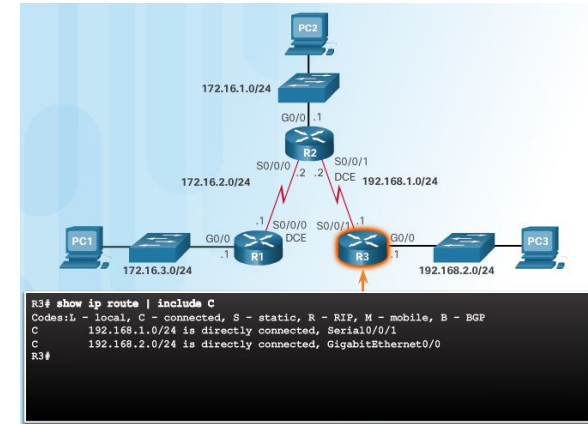
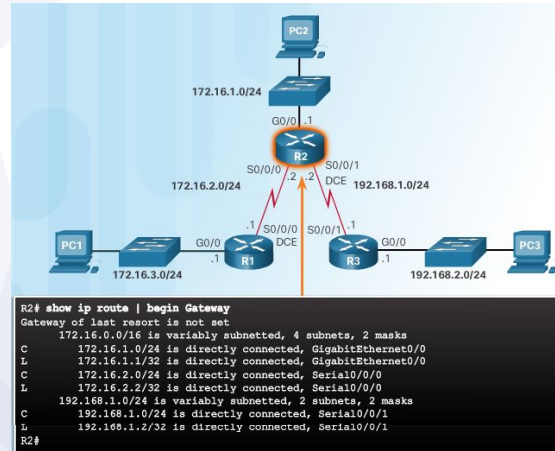
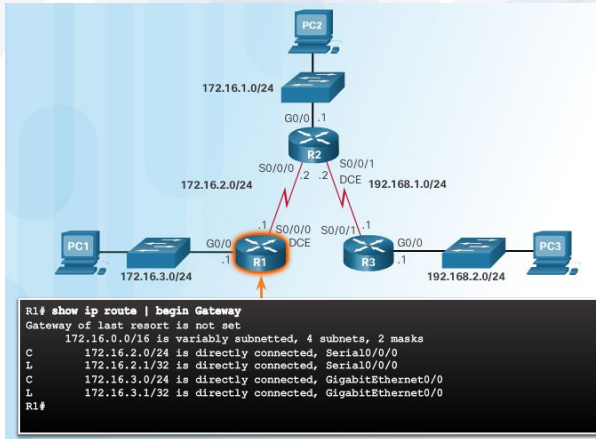
Parameter	Description
network-address	Destination network address of the remote network to be added to the routing table
subnet-mask	- Subnet mask of the remote network to be added to the routing table. - The subnet mask can be modified to summarize a group of networks.
ip-address	- Commonly referred to as the next-hop router's IP address. - Typically used when connecting to a broadcast media (i.e., Ethernet). - Commonly creates a recursive lookup
exit-intf	- Use the outgoing interface to forward packets to the destination network. - Also referred to as a directly attached static route. - Typically used when connecting in a point-to-point configuration.
distance	(Optional) Configures an administrative distance. - Typically used to configure a floating static route.

Configure IPv4 Static Routes: Next-Hop Options



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- In this example, each router only has entries for directly connected network



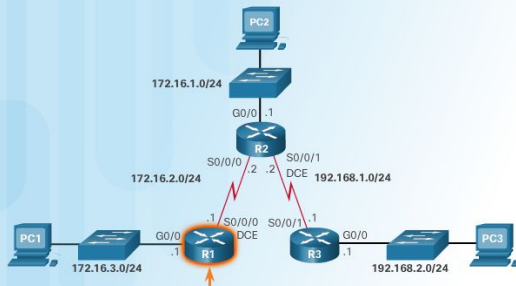
Configure IPv4 Static Routes: Next-Hop Options (Cont.)



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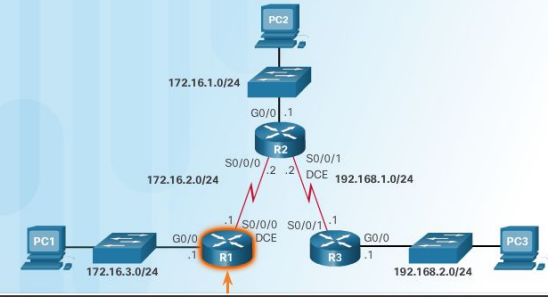
- R1 does not have an entry in its routing table for the R3 LAN network
- In a static route next-hop can be identified by
 - Next-hop IP address
 - Router exit interface
 - Next-hop IP address and exit interface

Verify Connectivity from R1 to R2



```
R1# ping 172.16.2.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.2.2, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 12/13/16 ms
R1#
```

Verify Connectivity from R1 to the R3 LAN



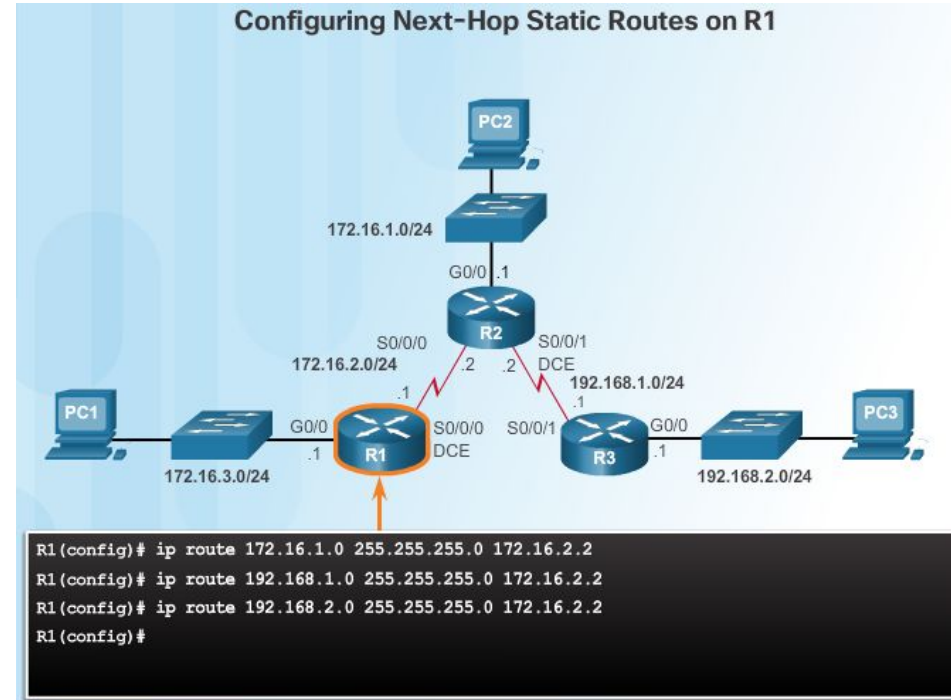
```
R1# ping 192.168.2.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.2.1, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
R1#
```

Configure IPv4 Static Routes: Configure a Next-Hop Static Route



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- In this example, only the next-hop IP address is specified
- Before packet is forwarded the router must determine the exit interface to use (route resolvability)

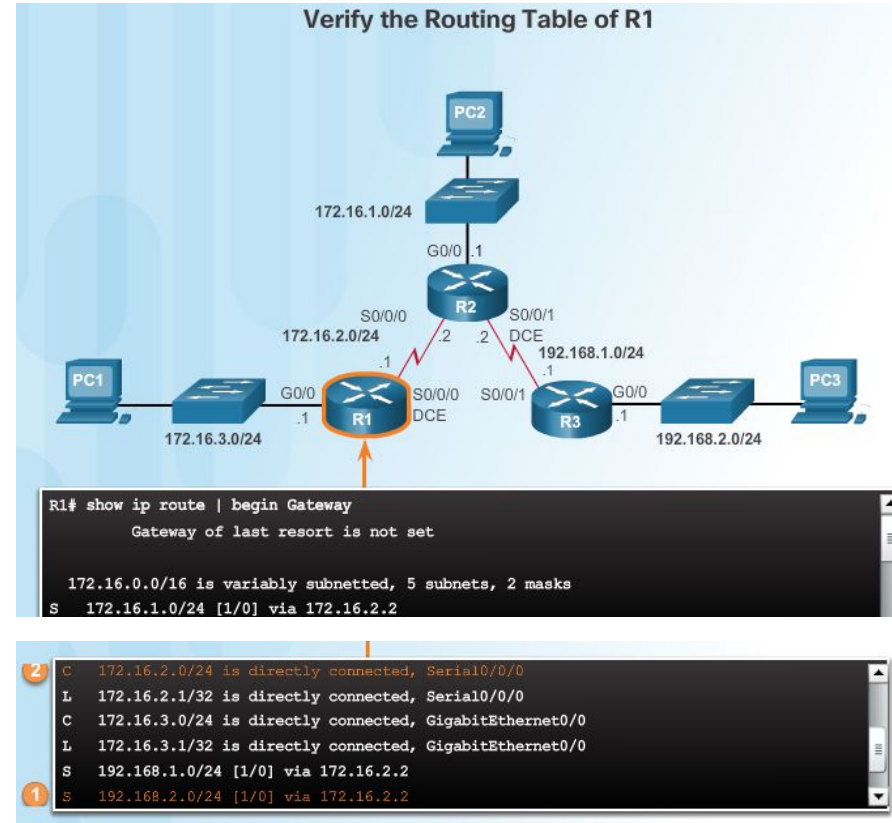


Configure IPv4 Static Routes: Configure a Next-Hop Static Route (Cont.)



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- In example, when a packet is destined for 192.168.2.0/24 network, R1:
 - Looks for match (#1) and needs to forward packets to 172.16.2.2
 - R1 must determine how to reach 172.16.2.2 first
 - Searches a second time for 172.16.2.0/24 (#2) and matches to exit interface s0/0/0
 - Takes two routing table lookups, process referred to as recursive lookup
 - If the exit interface is “down” or “administratively down” then the static route configured with next-hop will not be installed in routing table



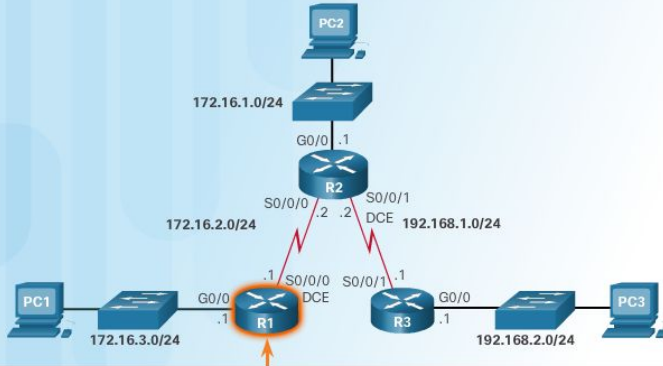
Configure IPv4 Static Routes: Configure a Directly Connected Static Route



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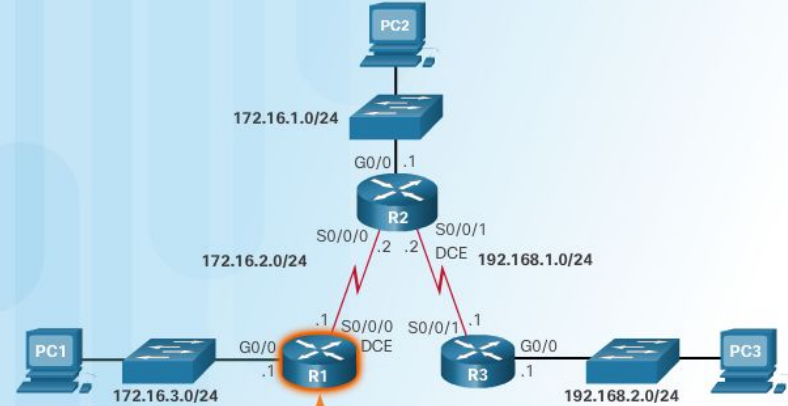
- Use the exit interface to specify next-hop so no other lookups are required
- Administrative distance of static route is 1

Configure Directly Connected Static Routes on R1



```
R1(config)# ip route 172.16.1.0 255.255.255.0 s0/0/0
R1(config)# ip route 192.168.1.0 255.255.255.0 s0/0/0
R1(config)# ip route 192.168.2.0 255.255.255.0 s0/0/0
R1(config)#
```

Verify the Routing Table of R1



```
R1# show ip route | begin Gateway
Gateway of last resort is not set

    172.16.0.0/16 is variably subnetted, 5 subnets, 2 masks
S       172.16.1.0/24 is directly connected, Serial0/0/0
C       172.16.2.0/24 is directly connected, Serial0/0/0
L       172.16.2.1/32 is directly connected, Serial0/0/0
C       172.16.3.0/24 is directly connected, GigabitEthernet0/0
L       172.16.3.1/32 is directly connected, GigabitEthernet0/0

S       192.168.1.0/24 is directly connected, Serial0/0/0
S       192.168.2.0/24 is directly connected, Serial0/0/0
R1#
```

Configure IPv4 Static Routes: Configure a Directly Connected Static Route (Cont.)



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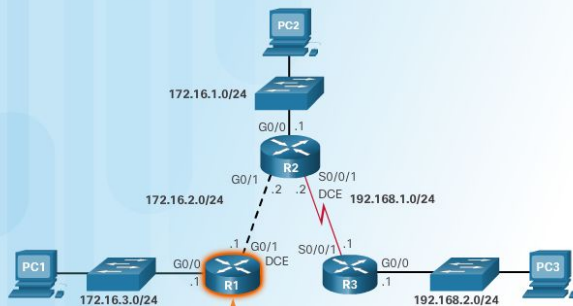
- Cisco Express Forwarding (CEF)
 - default behavior on IOS 12.0 or later
 - provides optimized lookup
 - uses a Forwarding Information Base (FIB) which is a copy of the routing table and an adjacency table that includes Layer 2 addresses
 - no recursive lookup needed for next-hop IP address lookups

Configure IPv4 Static Routes: Configure a Fully Specified Static Route



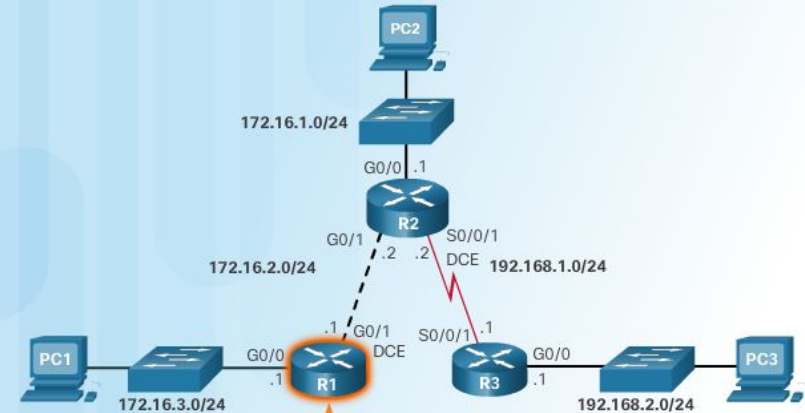
- Both the exit interface and the next-hop IP address are specified
- When exit interface is an Ethernet network, fully specified static route is used
- Note: With CEF, a next-hop address could be used instead

Configure Fully Specified Static Routes on R1



```
R1(config)# ip route 172.16.1.0 255.255.255.0 G0/1 172.16.2.2
R1(config)# ip route 192.168.1.0 255.255.255.0 G0/1 172.16.2.2
R1(config)# ip route 192.168.2.0 255.255.255.0 G0/1 172.16.2.2
R1(config)#
```

Verify the Routing Table of R1



```
R1# show ip route | begin Gateway
```

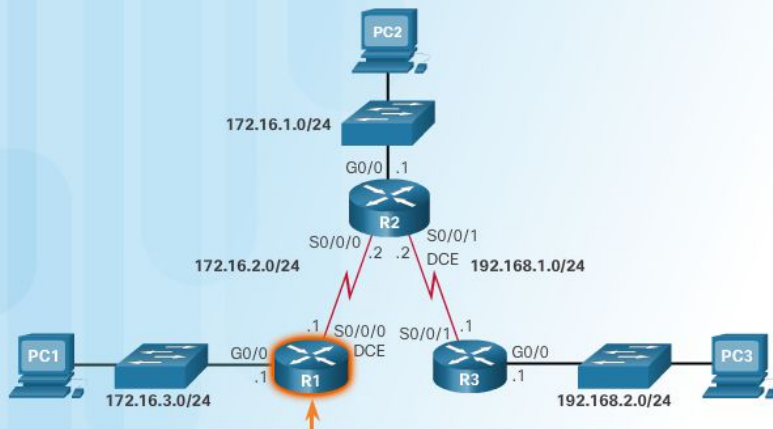
```
Gateway of last resort is not set
```

```
172.16.0.0/16 is variably subnetted, 5 subnets, 2 masks
S 172.16.1.0/24 [1/0] via 172.16.2.2, GigabitEthernet 0/1
C 172.16.2.0/24 is directly connected, GigabitEthernet 0/1
L 172.16.2.1/32 is directly connected, GigabitEthernet 0/1
C 172.16.3.0/24 is directly connected, GigabitEthernet0/0
L 172.16.3.1/32 is directly connected, GigabitEthernet0/0
S 192.168.1.0/24 [1/0] via 172.16.2.2, GigabitEthernet 0/1
S 192.168.2.0/24 [1/0] via 172.16.2.2, GigabitEthernet 0/1
R1#
```

Configure IPv4 Static Routes: Verify a Static Route



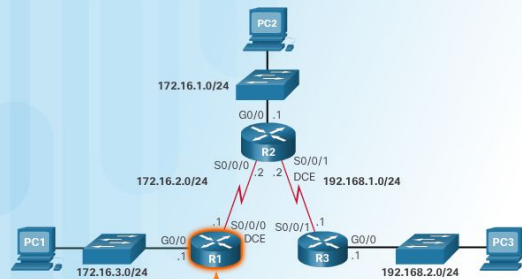
Verify the Routing Table of R1



```
R1# show ip route static | begin Gateway
Gateway of last resort is not set
```

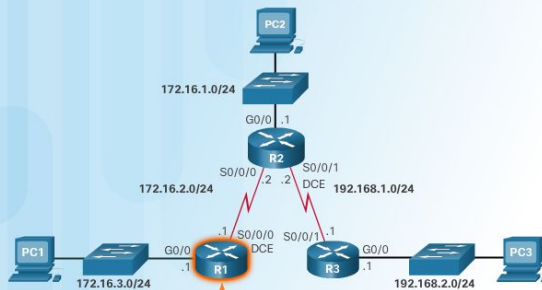
```
S      172.16.0.0/16 is variably subnetted, 5 subnets, 2 masks
S      172.16.1.0/24 [1/0] via 172.16.2.2
S      192.168.1.0/24 [1/0] via 172.16.2.2
S      192.168.2.0/24 [1/0] via 172.16.2.2
R1#
```

Verify a Specific Entry in the Routing Table



```
R1# show ip route 192.168.2.1
Routing entry for 192.168.2.0/24
Known via "static", distance 1, metric 0
Routing Descriptor Blocks:
* 172.16.2.2
  Route metric is 0, traffic share count is 1
R1#
```

Verify the Static Route Configuration



```
R1# show running-config | section ip route
ip route 172.16.1.0 255.255.255.0 172.16.2.2
ip route 192.168.1.0 255.255.255.0 172.16.2.2
ip route 192.168.2.0 255.255.255.0 172.16.2.2
R1#
```

Configure IPv4 Default Routes: Default Static Route



- Default static routes are commonly used when connecting:
 - An edge router to a service provider network
 - A stub router (a router with only one upstream neighbor router)
- Default route is used when no other routes in the routing table match the destination IP

Default Static Route Syntax

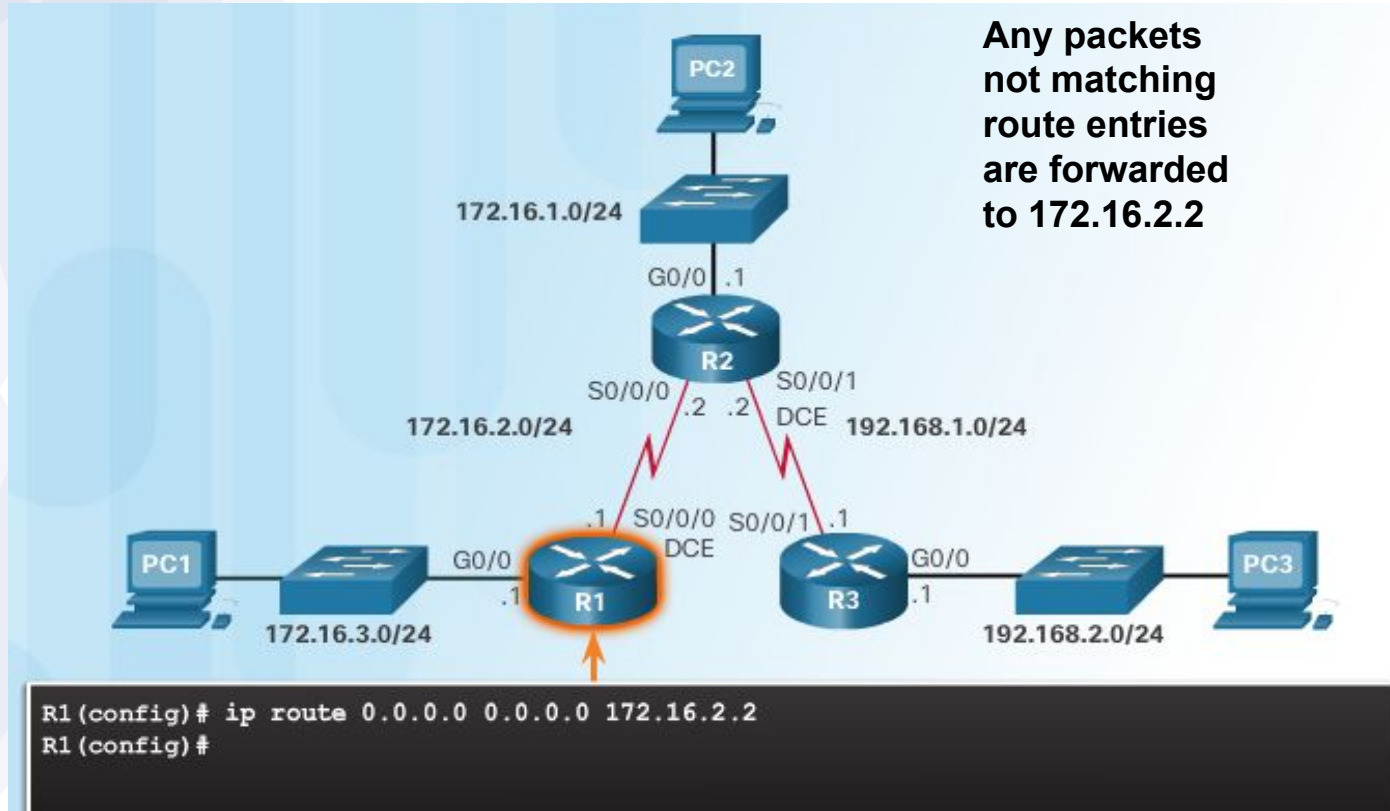
```
Router(config)#ip route 0.0.0.0 0.0.0.0 {ip-address | exit-intf}
```

Parameter	Description
0.0.0.0 0.0.0.0	Matches any network address.
ip-address	• Commonly referred to as the next-hop router's IP address. • Typically used when connecting to a broadcast media (i.e., Ethernet). • Commonly creates a recursive lookup.
exit-intf	• Use the outgoing interface to forward packets to the destination network. • Also referred to as a directly attached static route. • Typically used when connecting in a point-to-point configuration.

Configure IPv4 Default Routes: Configure a Default Static Route



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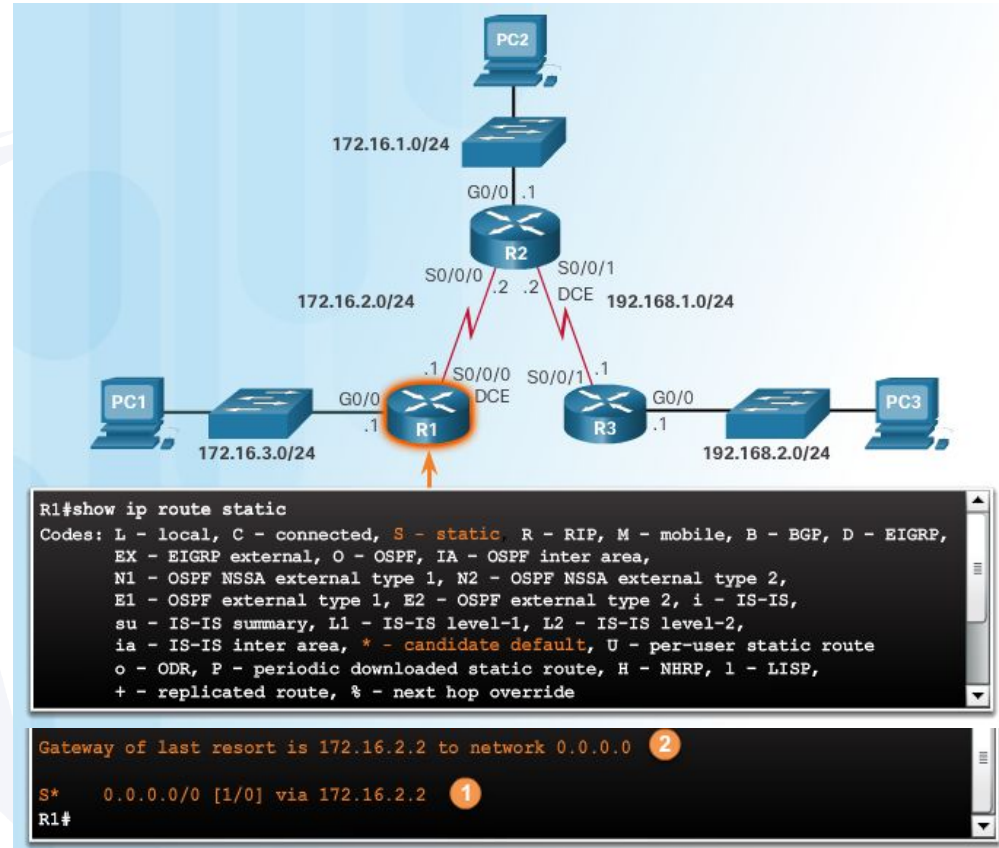


Configure IPv4 Default Routes: Verify a Default Static Route



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- **show ip route static**
displays just the static routes
 - S indicates static route
 - candidate default route indicated by *
 - /0 mask in route entry indicates none of the bits are required to match



Configure IPv6 Static Routes: The ipv6 route Command



```
Router(config)# ipv6 route ipv6-prefix/prefix-length {ipv6-address | exit-intf}
```

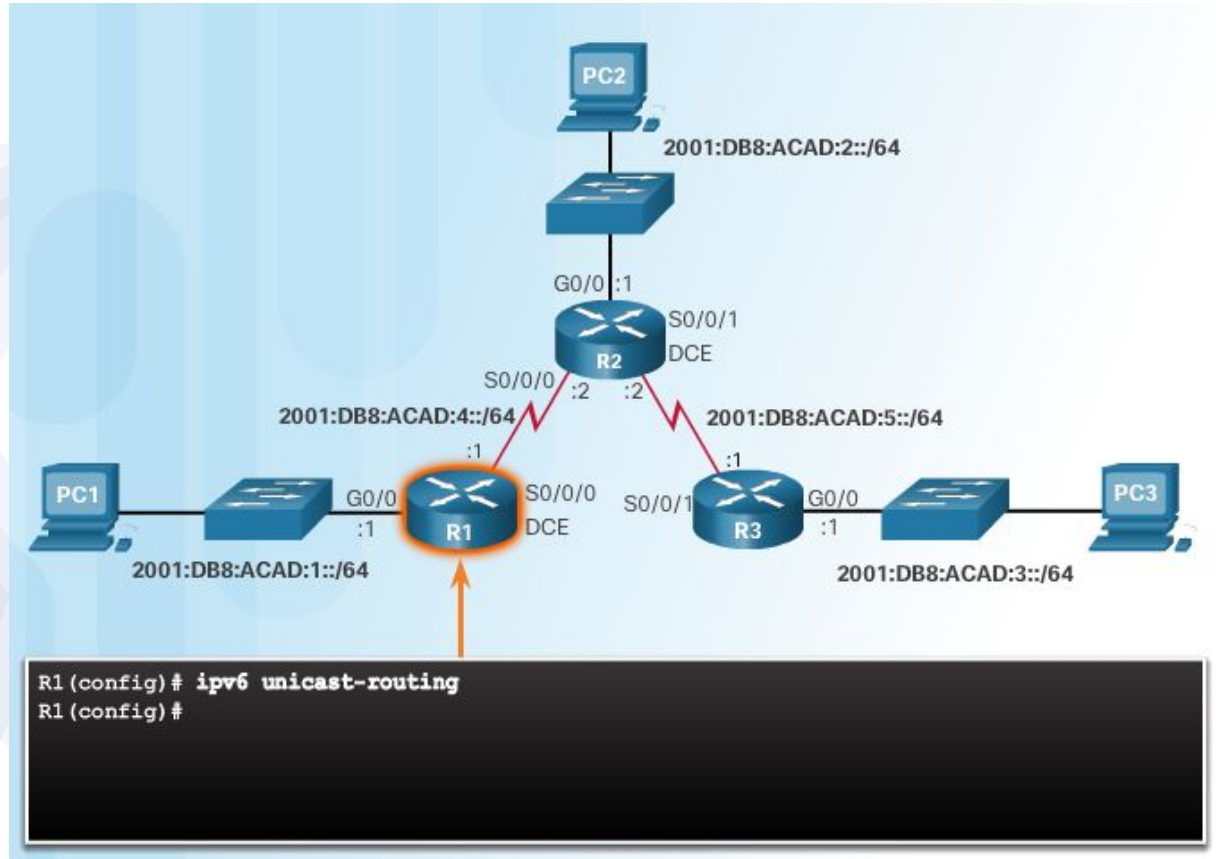
Parameter	Description
ipv6-prefix	Destination network address of the remote network to be added to the routing table.
prefix-length	Prefix length of the remote network to be added to the routing table.
ipv6-address	• Commonly referred to as the next-hop router's IP address. • Typically used when connecting to a broadcast media (i.e., Ethernet). • Commonly creates a recursive lookup.
exit-intf	• Use the outgoing interface to forward packets to the destination network. • Also referred to as a directly attached static route. • Typically used when connecting in a point-to-point configuration.

Configure IPv6 Static Routes: The ipv6 route Command (Cont.)



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- **ipv6 unicast-routing** enables the router to forward IPv6 packets

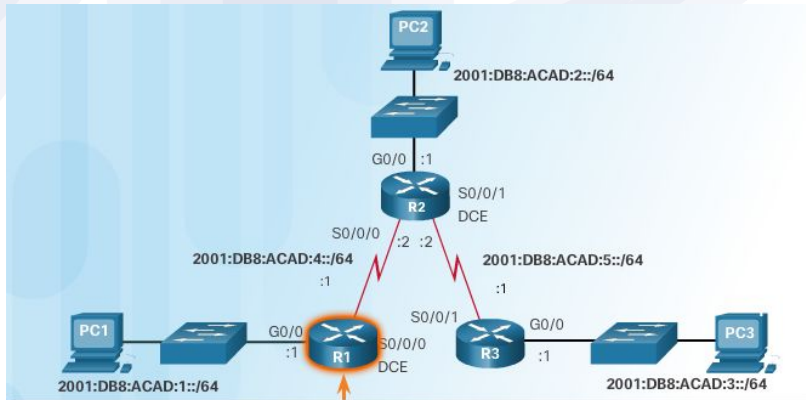


Configure IPv6 Static Routes: Next-Hop Options



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- Each router only knows about directly connected networks
 - R1 can **ping** R2 (ipv6 2001:DB8:ACAD:4::2) but cannot **ping** R3 (ipv6 2001:DB8:ACAD:3::2)



```
R1#show ipv6 route
<output omitted>
C 2001:DB8:ACAD:1::/64 [0/0]
  via GigabitEthernet0/0, directly connected
L 2001:DB8:ACAD:1::1/128 [0/0]
  via GigabitEthernet0/0, receive
C 2001:DB8:ACAD:4::/64 [0/0]
  via Serial0/0/0, directly connected
L 2001:DB8:ACAD:4::1/128 [0/0]
  via Serial0/0/0, receive
L FF00::/8 [0/0]
  via Null0, receive
R1#
```

```
R2#show ipv6 route
<output omitted>
C 2001:DB8:ACAD:2::/64 [0/0]
  via GigabitEthernet0/0, directly connected
L 2001:DB8:ACAD:2::1/128 [0/0]
  via GigabitEthernet0/0, receive
C 2001:DB8:ACAD:4::/64 [0/0]
```

```
  via Serial0/0/0, directly connected
L 2001:DB8:ACAD:4::2/128 [0/0]
  via Serial0/0/0, receive
C 2001:DB8:ACAD:5::/64 [0/0]
  via Serial0/0/1, directly connected
L 2001:DB8:ACAD:5::2/128 [0/0]
  via Serial0/0/1, receive
L FF00::/8 [0/0]
  via Null0, receive
R2#
```

```
R3#show ipv6 route
<output omitted>
C 2001:DB8:ACAD:3::/64 [0/0]
  via GigabitEthernet0/0, directly connected
L 2001:DB8:ACAD:3::1/128 [0/0]
  via GigabitEthernet0/0, receive
C 2001:DB8:ACAD:5::/64 [0/0]
  via Serial0/0/1, directly connected
L 2001:DB8:ACAD:5::1/128 [0/0]
  via Serial0/0/1, receive
L FF00::/8 [0/0]
  via Null0, receive
R3#
```

Configure IPv6 Static Routes: Next-Hop Options (Cont.)



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- Next hop can be identified by an IPv6 address, exit interface, or both.
- Destination is specified by one of three route types:
 - **Next-hop static IPv6 route** - Only the next-hop IPv6 address is specified
 - **Directly connected static IPv6 route** - Only the router exit interface is specified
 - **Fully specified static IPv6 route** - The next-hop IPv6 address and exit interface are specified

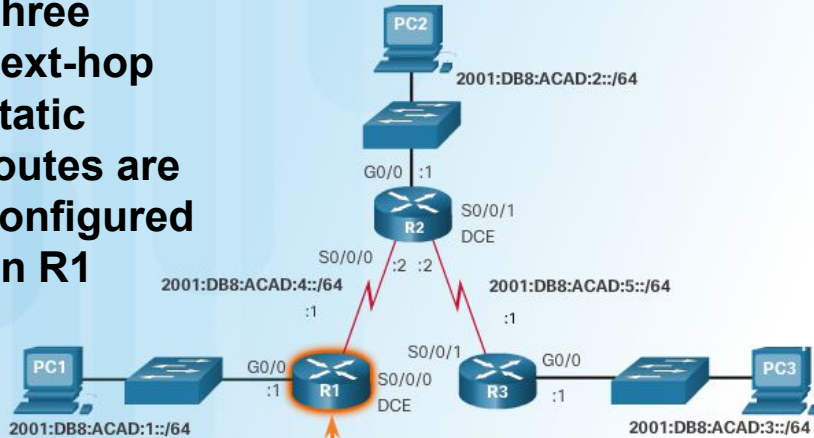
Configure IPv6 Static Routes: Configure a Next Hop Static IPv6 Route



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As with IPv4, must resolve the route to determine the exit interface to use to forward the packet

Three next-hop static routes are configured on R1



```
R1(config)# ipv6 route 2001:DB8:ACAD:2::/64 2001:DB8:ACAD:4::2
R1(config)# ipv6 route 2001:DB8:ACAD:5::/64 2001:DB8:ACAD:4::2
R1(config)# ipv6 route 2001:DB8:ACAD:3::/64 2001:DB8:ACAD:4::2
R1(config)#
```

```
R1# show ipv6 route
IPv6 Routing Table - default - 8 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route, B - BGP,
       R - RIP, H - NHRP, I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea,
       IS - ISIS summary, D - EIGRP, EX - EIGRP external, ND - ND Default,
       NDp - ND Prefix, DCE - Destination, Ndr - Redirect, O - OSPF Intra,
       OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1,
       ON2 - OSPF NSSA ext 2

C 2001:DB8:ACAD:1::/64 [0/0]
  via GigabitEthernet0/0, directly connected
L 2001:DB8:ACAD:1::1/128 [0/0]
  via GigabitEthernet0/0, receive
S 2001:DB8:ACAD:2::/64 [1/0]
  via 2001:DB8:ACAD:4::2
S 2001:DB8:ACAD:3::/64 [1/0]
  via 2001:DB8:ACAD:4::2
C 2001:DB8:ACAD:4::/64 [0/0]
  via Serial0/0/0, directly connected
L 2001:DB8:ACAD:4::1/128 [0/0]
  via Serial0/0/0, receive
S 2001:DB8:ACAD:5::/64 [1/0]
  via 2001:DB8:ACAD:4::2
L FF00::/8 [0/0]
  via Null0, receive
R1#
```

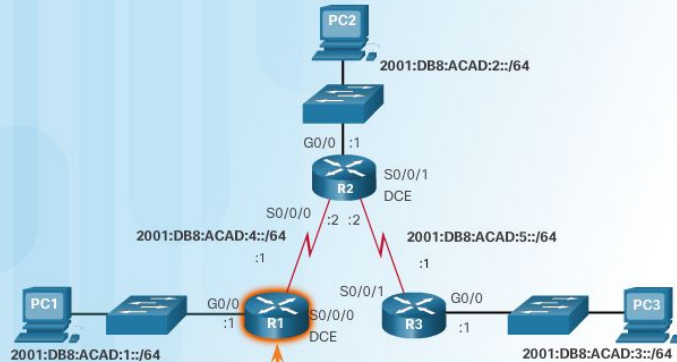
The IPv6 address matches the route for the directly connected network 2001:DB8:ACAD:4::/64 with the exit interface Serial 0/0/0.

Configure IPv6 Static Routes: Configure a Directly Connected Static IPv6 Route



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- Alternative to next hop is to specify the exit interface
- Packet destined for 2001:DB8:ACAD:3::/64 network, forwarded out Serial 0/0/0 – no other lookups needed



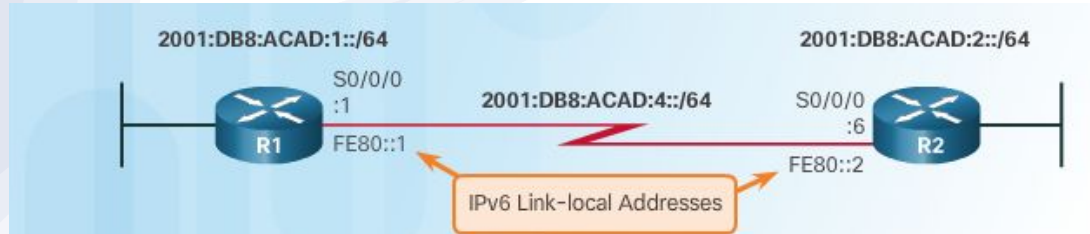
```
R1(config)# ipv6 route 2001:DB8:ACAD:2::/64 s0/0/0
R1(config)# ipv6 route 2001:DB8:ACAD:5::/64 s0/0/0
R1(config)# ipv6 route 2001:DB8:ACAD:3::/64 s0/0/0
R1(config)#
R1#
```

```
R1# show ipv6 route
IPv6 Routing Table - default - 8 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
       B - BGP, R - RIP, I1 - ISIS L1, I2 - ISIS L2
       IA - ISIS interarea, IS - ISIS summary, D - EIGRP, EX - EIGRP external
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
       O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
C    2001:DB8:ACAD:1::/64 [0/0]
    via GigabitEthernet0/0, directly connected
L    2001:DB8:ACAD:1::1/128 [0/0]
    via GigabitEthernet0/0, receive
S    2001:DB8:ACAD:2::/64 [1/0]
    via Serial0/0/0, directly connected
S    2001:DB8:ACAD:3::/64 [1/0]
    via Serial0/0/0, directly connected
C    2001:DB8:ACAD:4::/64 [0/0]
    via Serial0/0/0, directly connected
L    2001:DB8:ACAD:4::1/128 [0/0]
    via Serial0/0/0, receive
S    2001:DB8:ACAD:5::/64 [1/0]
    via Serial0/0/0, directly connected
L    FF00::/8 [0/0]
    via Null0, receive
R1#
```

Configure IPv6 Static Routes: Configure a Fully Specified Static IPv6 Route



- Fully specified static route must be used if IPv6 link-local address is used as next-hop



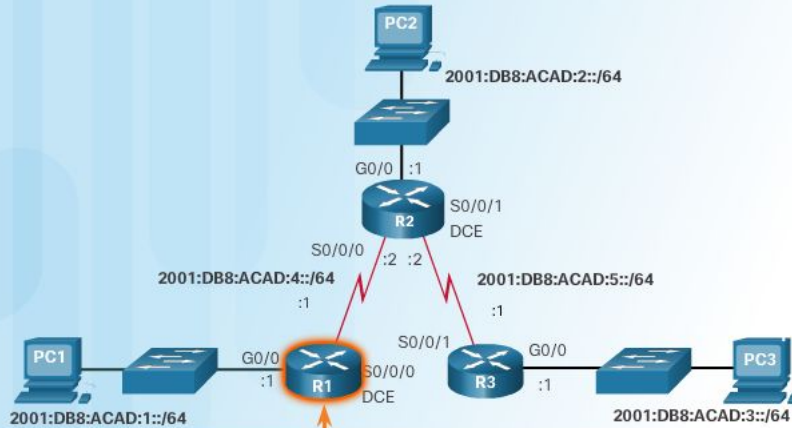
```
R1(config)# ipv6 route 2001:db8:acad:2::/64 fe80::2
% Interface has to be specified for a link-local nexthop
R1(config)# ipv6 route 2001:db8:acad:2::/64 s0/0/0 fe80::2
R1(config)#
```

```
R1# show ipv6 route static | begin 2001:DB8:ACAD:2::/64
S   2001:DB8:ACAD:2::/64 [1/0]
    via FE80::2, Serial10/0/0
```

Configure IPv6 Static Routes: Verify IPv6 Static Routes



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```
R1# show ipv6 route static
IPv6 Routing Table - default - 8 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route, B - BGP,
R - RIP, I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary,
D - EIGRP, EX - EIGRP external, ND - ND Default, NDp - ND Prefix,
DCE - Destination, NDr - Redirect, O - OSPF Intra, OI - OSPF Inter,
OE1 - OSPF ext 1, OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1,
ON2 - OSPF NSSA ext 2
S 2001:DB8:ACAD:2::/64 [1/0]
  via 2001:DB8:ACAD:4::2
```

```
S 2001:DB8:ACAD:3::/64 [1/0]
  via 2001:DB8:ACAD:4::2
S 2001:DB8:ACAD:5::/64 [1/0]
  via 2001:DB8:ACAD:4::2
R1#
```

```
R1# show ipv6 route 2001:db8:acad:3::
Routing entry for 2001:DB8:ACAD:3::/64
Known via "static", distance 1, metric 0
Route count is 1/1, share count 0
Routing paths:
  2001:DB8:ACAD:4::2
    Last updated 00:19:11 ago
R1#
```

```
R1# show running-config | section ipv6 route
ipv6 route 2001:DB8:ACAD:2::/64 2001:DB8:ACAD:4::2
ipv6 route 2001:DB8:ACAD:3::/64 2001:DB8:ACAD:4::2
ipv6 route 2001:DB8:ACAD:5::/64 2001:DB8:ACAD:4::2
R1#
```

Configure IPv6 Default Routes: Default Static IPv6 Route

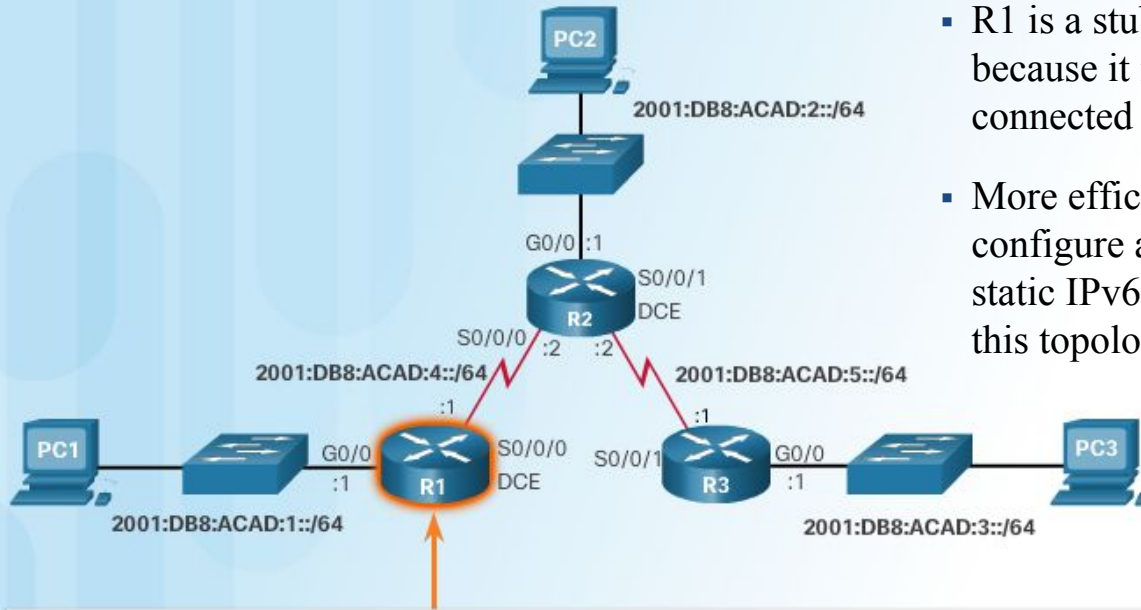


- Default static route matches all packets not specified in routing table

```
Router(config)# ipv6 route ::/0 (ipv6-address | exit-intf)
```

Parameter	Description
::/0	Matches any IPv6 prefix regardless of prefix length.
ipv6-address	• Commonly referred to as the next-hop router's IPv6 address. • Typically used when connecting to a broadcast media (i.e., Ethernet). • Commonly creates a recursive lookup.
exit-intf	• Use the outgoing interface to forward packets to the destination network. • Also referred to as a directly attached static route. • Typically used when connecting in a point-to-point configuration.

- R1 is a stub router because it is only connected to R2
- More efficient to configure a default static IPv6 route in this topology



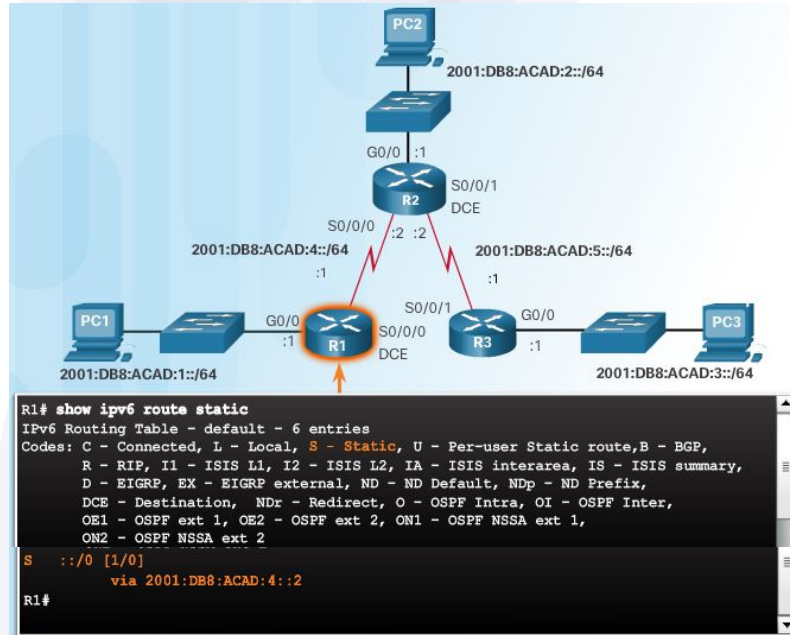
```
R1(config)# ipv6 route ::/0 2001:DB8:ACAD:4::2
R1(config)#
```

Configure IPv6 Default Routes: Verify a Default Static Route



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- `::/0` mask indicates that none of the bits are required to match
- If a more specific match does not exist, the default static IPv6 route matches all packets.



```
R1# ping 2001:0DB8:ACAD:3::1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:ACAD:3::1,
timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max
= 28/28/28 ms
R1#
```

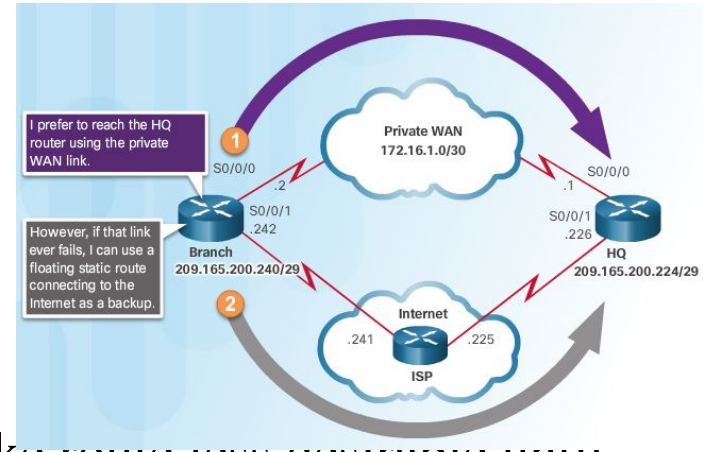
Configure Floating Static Routes: Floating Static Routes



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Floating static routes have an administrative distance greater than the dynamic routing protocol or other static route

- Used as backup routes
- Administrative distance of common routing protocols
 - EIGRP = 90
 - IGRP = 100
 - OSPF = 110
 - IS-IS = 115
 - RIP = 120
- By default, AD of static route = 1
- Static route AD can be increased to make route less desirable until preferred route is lost

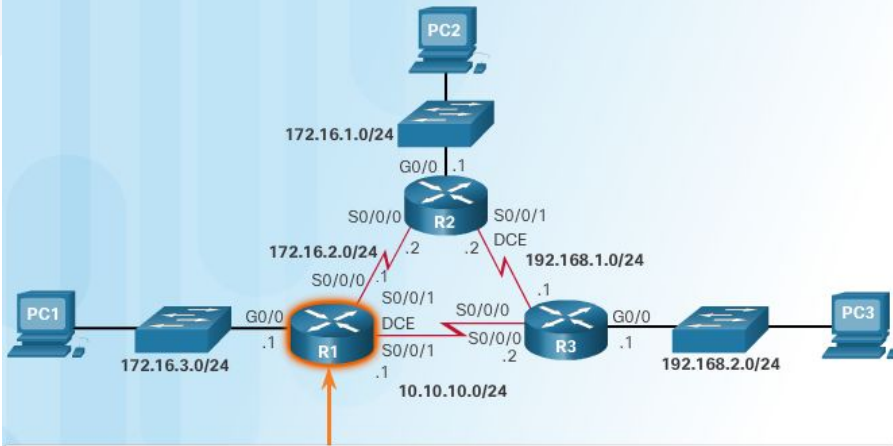


Configure Floating Static Routes: Configure a Floating Static Route



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Configuring a Floating Static Route to R3



```
R1(config)# ip route 0.0.0.0 0.0.0.0 172.16.2.2
R1(config)# ip route 0.0.0.0 0.0.0.0 10.10.10.2 5
R1(config)#
```

Preferred router from
R1 is to R2 (AD = 1)

```
R1# show ip route static | begin Gateway
Gateway of last resort is 0.0.0.0 to network 0.0.0.0

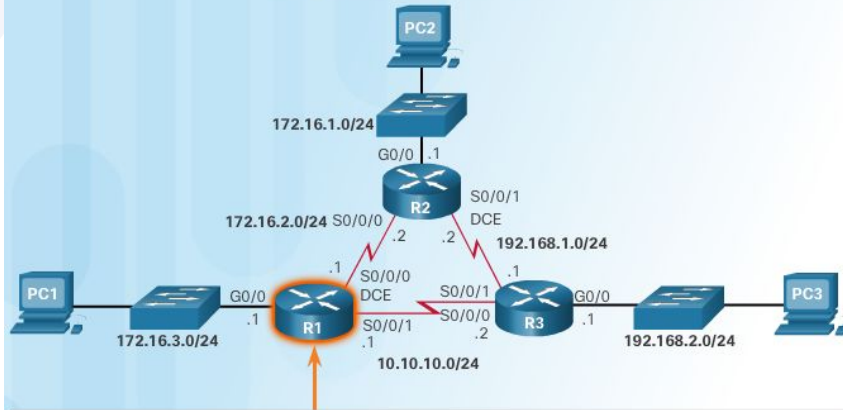
S* 0.0.0.0/0 [1/0] via 172.16.2.2
R1#
```

Configure Floating Static Routes: Test the IPv4 Floating Static Route



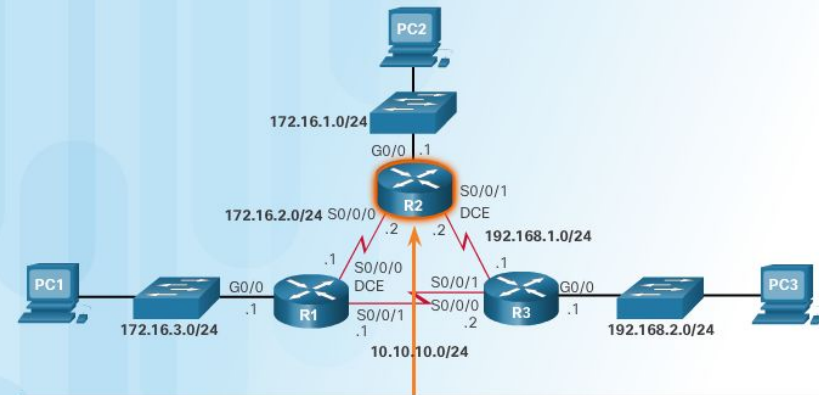
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Verify the Path to the R3 LAN



```
R1# traceroute 192.168.2.1
Type escape sequence to abort.
Tracing the route to 192.168.2.1
VRF info: (vrf in name/id, vrf out name/id)
 1 172.16.2.2 4 msec 4 msec 8 msec
 2 192.168.1.1 12 msec * 12 msec
R1#
```

Simulate a Router Failure on R2



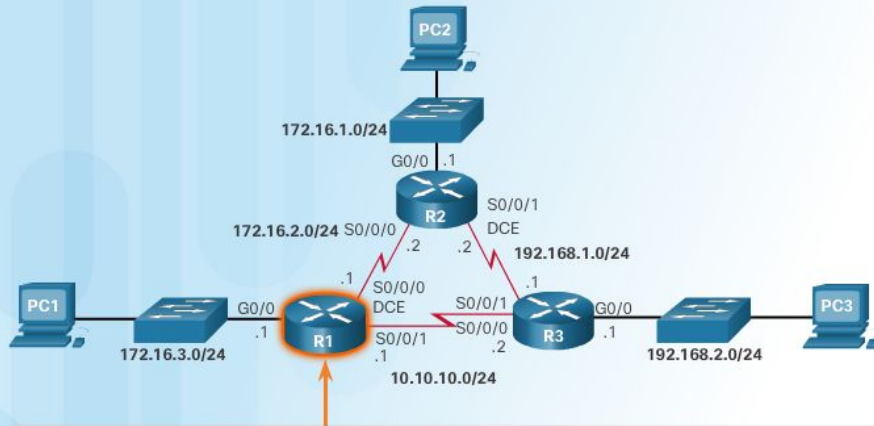
```
R2(config)# int s0/0/0
R2(config-if)# shut
*Feb 21 16:33:35.939: %LINK-5-CHANGED: Interface Serial0/0/0, changed
state to administratively down
*Feb 21 16:33:36.939: %LINEPROTO-5-UPDOWN: Line protocol on Interface
Serial0/0/0, changed state to down
R2(config-if)# int s0/0/1
R2(config-if)# shut
R2(config-if)#
*Feb 21 16:33:42.543: %LINK-5-CHANGED: Interface Serial0/0/1, changed
```

Configure Floating Static Routes: Test the IPv4 Floating Static Route (Cont.)



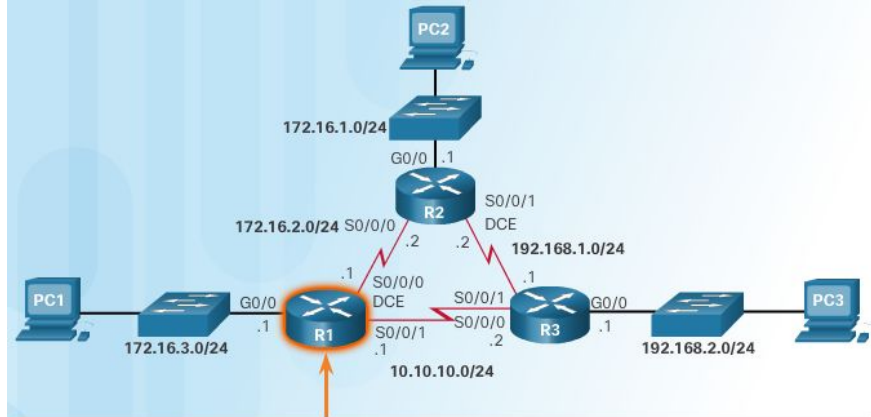
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Verify the Default Route on R1



```
*Feb 21 16:35:58.435: %LINK-3-UPDOWN: Interface Serial0/0/0, changed
state to down
*Feb 21 16:35:59.435: %LINEPROTO-5-UPDOWN: Line protocol on Interface
Serial0/0/0, changed state to down
R1#
R1# show ip route static | begin Gateway
Gateway of last resort is 0.0.0.0 to network 0.0.0.0
S* 0.0.0.0/0 [5/0] via 10.10.10.2
R1#
```

Verify the Path to the R3 LAN



```
R1# traceroute 192.168.2.1
Type escape sequence to abort.
Tracing the route to 192.168.2.1
VRF info: (vrf in name/id, vrf out name/id)
 1 10.10.10.2 4 msec 4 msec *
R1#
```

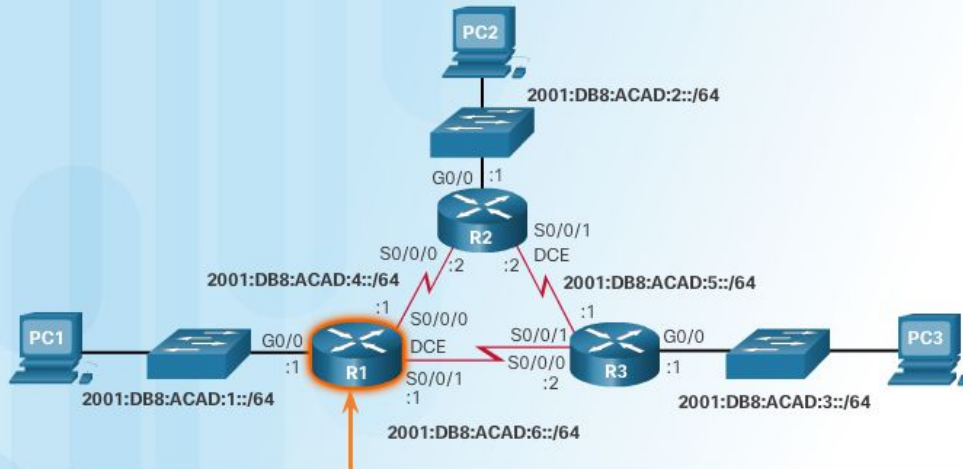
Configure Floating Static Routes: Configure an IPv6 Floating Static Route



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- Similar to IPv4 floating static routes

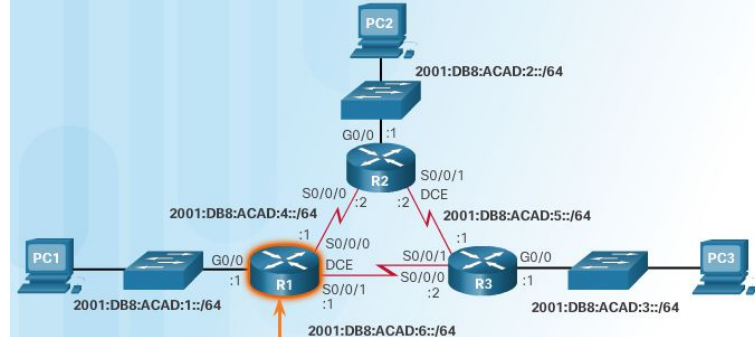
Configure an IPv6 Floating Static Routes to R3



```
R1(config)# ipv6 route ::/0 2001:db8:acad:4::2  
R1(config)# ipv6 route ::/0 2001:db8:acad:6::2 5  
R1(config)#
```

```
R1# show run | include ipv6 route  
ipv6 route ::/0 2001:DB8:ACAD:6::2 5  
ipv6 route ::/0 2001:DB8:ACAD:4::2  
R1#
```

Verify the IPv6 Floating Static Route is not in the Routing Table



```
R1# show ipv6 route static | begin S :  
S ::/0 [1/0]  
via 2001:DB8:ACAD:4::2  
R1#
```

Configure Static Host Routes: Automatically Installed Host Routes



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Host route is an IPv4 address with a 32-bit mask or IPv6 address with a 128-bit mask.

- Automatically installed when IP address is configured
- Configured as a static host route
- Allows more efficiency for packets directed to the router
- Local route marked with “L” (introduced in IOS 15)

Branch IPv4 Routing Table



```
Branch# show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, I - LISP
a - application route
+ - replicated route, % - next hop override
Gateway of last resort is not set

198.51.100.0/24 is variably subnetted, 2 subnets, 2 masks
C    198.51.100.0/30 is directly connected, Serial0/0/0
L    198.51.100.1/32 is directly connected, Serial0/0/0
Branch#
```

Branch IPv6 Routing Table



```
Branch# show ipv6 route
IPv6 Routing Table - default - 3 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
B - BGP, R - RIP, H - NHRP, I1 - ISIS L1
I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary, D - EIGRP
EX - EIGRP external, ND - ND Default, NDp - ND Prefix, DCE - Destination
NDR - Redirect, O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1
OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
a - Application

C    2001:DB8:ACAD:1::/64 [0/0]
    via Serial0/0/0, directly connected
L    2001:DB8:ACAD:1::1/128 [0/0]
    via Serial0/0/0, receive
L    FF00::/8 [0/0]
    via Null0, receive
Branch#
```

Configure Static Host Routes: Configure IPv4 and IPv6 Static Host Routes



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Fully Specified IPv6 Host Route with the Next-Hop Link-Local Address



```
Branch(config)# no ipv6 route 2001:db8:acad:2::99/128 2001:db8:acad:1::2
Branch(config)# ipv6 route 2001:db8:acad:2::99/128 serial 0/0/0 fe80::2
Branch(config)# end
Branch# show ipv6 route
```

```
S ::/0 [1/0]
  via 2001:DB8:ACAD:1::2
C 2001:DB8:ACAD:1::/64 [0/0]
  via Serial0/0/0, directly connected
L 2001:DB8:ACAD:1::1/128 [0/0]
  via Serial0/0/0, receive
S 2001:DB8:ACAD:2::99/128 [1/0]
  via FE80::2, Serial0/0/0
L FF00::/8 [0/0]
  via Null0, receive
Branch#
```

IPv4 and IPv6 Host Route Configuration and Verification



```
Branch(config)# ip route 209.165.200.238 255.255.255.255 198.51.100.2
Branch(config)# ipv6 route 2001:db8:acad:2::99/128 2001:db8:acad:1::2
Branch(config)# end
Branch# show ip route | begin Gateway
Gateway of last resort is not set
```

```
198.51.100.0/24 is variably subnetted, 2 subnets, 2 masks
C 198.51.100.0/30 is directly connected, Serial0/0/0
L 198.51.100.1/32 is directly connected, Serial0/0/0
209.165.200.0/32 is subnetted, 1 subnets
S 209.165.200.238 [1/0] via 198.51.100.2
Branch# show ipv6 route
```

```
C 2001:DB8:ACAD:1::/64 [0/0]
  via Serial0/0/0, directly connected
L 2001:DB8:ACAD:1::1/128 [0/0]
  via Serial0/0/0, receive
S 2001:DB8:ACAD:2::99/128 [1/0]
  via 2001:DB8:ACAD:1::2
L FF00::/8 [0/0]
  via Null0, receive
Branch#
```



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Troubleshoot Static and Default Routes

Packet Processing with Static Routes

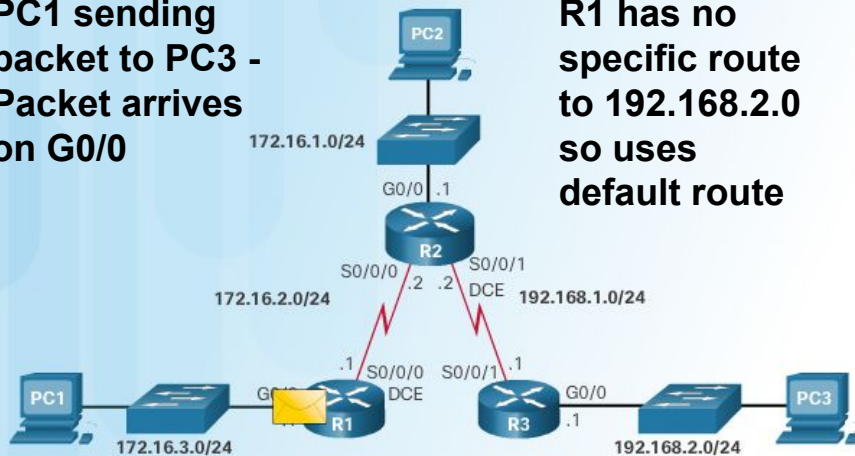
Static Routes and Packet Forwarding



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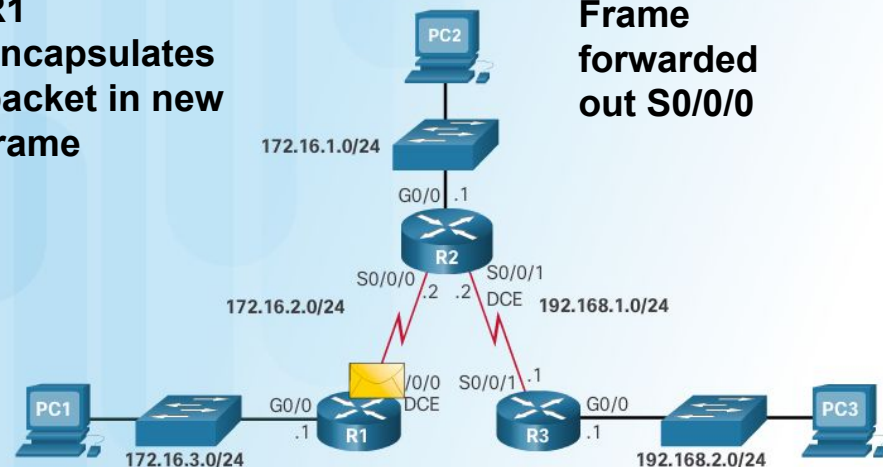
**PC1 sending
packet to PC3 -
Packet arrives
on G0/0**

**R1 has no
specific route
to 192.168.2.0
so uses
default route**



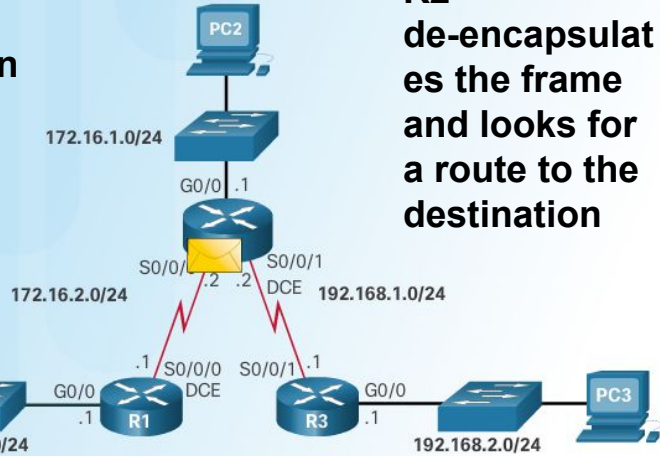
**R1
encapsulates
packet in new
frame**

**Frame
forwarded
out S0/0/0**



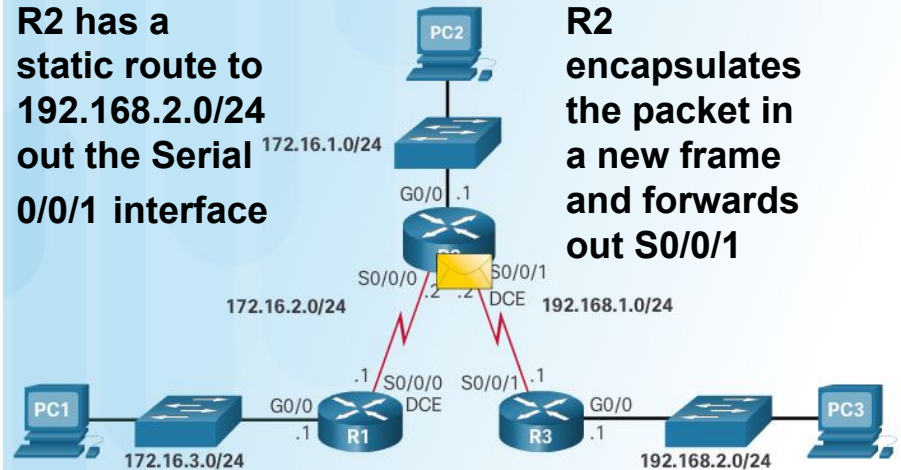


**Packet
arrives on
S0/0/0
interface
on R2**



**R2
de-encapsulat
es the frame
and looks for
a route to the
destination**

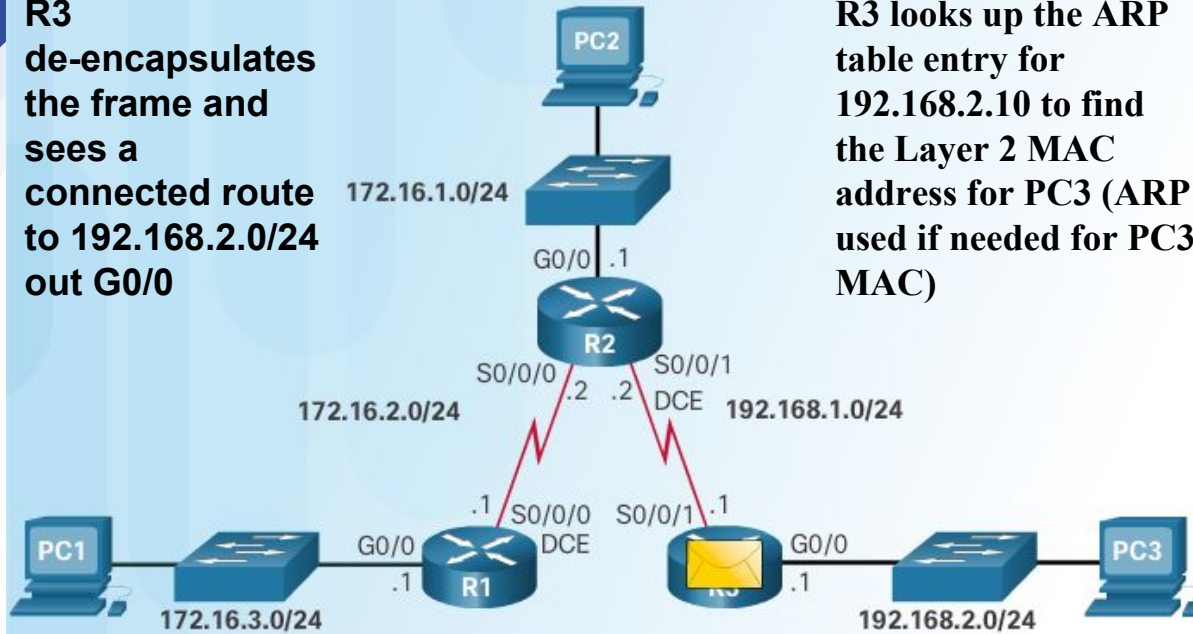
**R2 has a
static route to
192.168.2.0/24
out the Serial
0/0/1 interface**



**R2
encapsulates
the packet in
a new frame
and forwards
out S0/0/1**

**R3
de-encapsulates
the frame and
sees a
connected route
to 192.168.2.0/24
out G0/0**

**R3 looks up the ARP
table entry for
192.168.2.10 to find
the Layer 2 MAC
address for PC3 (ARP
used if needed for PC3
MAC)**



- R3 encapsulates the packet in a new frame with the MAC address of the G0/0 interface as the source Layer 2 address and the MAC address of PC3 as the destination MAC address
- Frame is forwarded out of G0/0 interface and packet arrives on the NIC interface of PC3

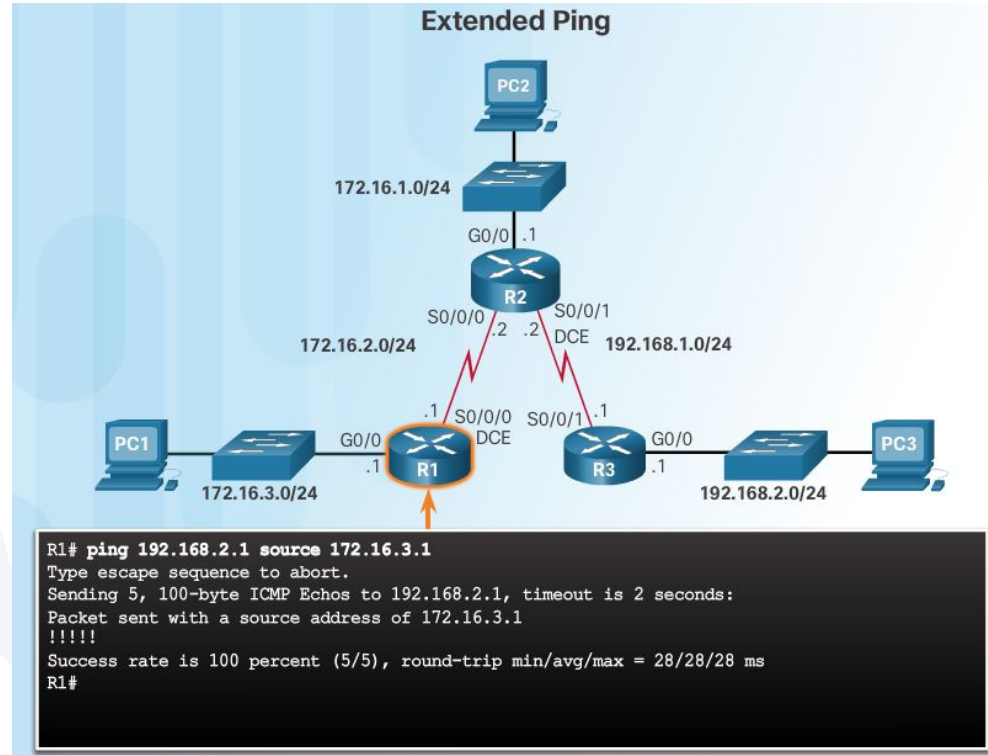
Troubleshoot IPv4 Static and Default Route Configuration

Troubleshoot a Missing Route



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- Common IOS troubleshooting commands include:
 - ping
 - traceroute
 - show ip route
 - show ip interface brief
 - show cdp neighbors detail

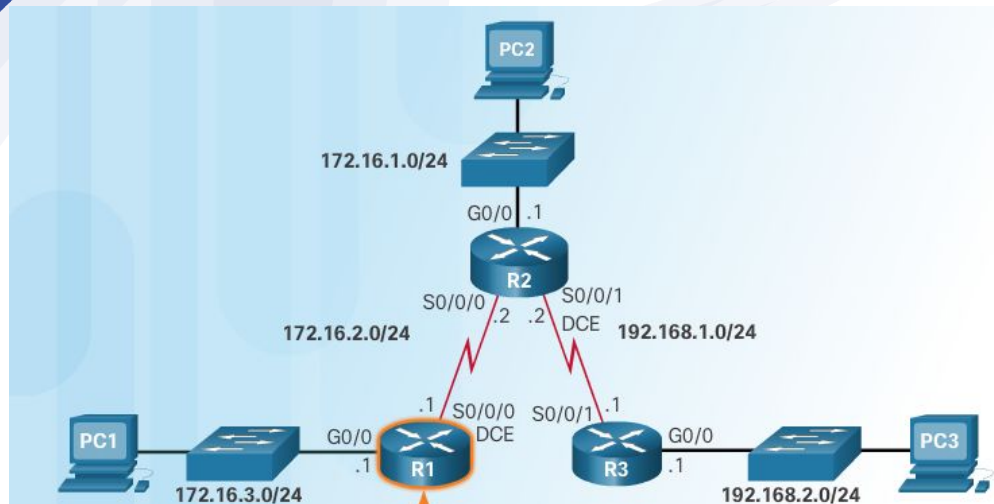


Troubleshoot IPv4 Static and Default Route Configuration

Troubleshoot a Missing Route (Cont.)



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```
R1# show ip interface brief
Interface      IP-Address OK? Method Status      Protocol
Embedded-Service-Engine0/0 unassigned YES unset  administratively down down
GigabitEthernet0/0 172.16.3.1 YES manual up          up
GigabitEthernet0/1 unassigned YES unset  administratively down down
Serial0/0/0      172.16.2.1 YES manual up          up
Serial0/0/1      unassigned YES unset  administratively down down
R1#
```

```
R1# traceroute 192.168.2.1
Type escape sequence to abort.
Tracing the route to 192.168.2.1
VRF info: (vrf in name/id, vrf out name/id)
 1 172.16.2.2 4 msec 4 msec 8 msec
 2 192.168.1.1 12 msec 12 msec *
```

```
R1# show ip route | begin Gateway
Gateway of last resort is not set
```

```

      172.16.0.0/16 is variably subnetted, 5 subnets, 2 masks
S       172.16.1.0/24 [1/0] via 172.16.2.2
C       172.16.2.0/24 is directly connected, Serial0/0/0
L       172.16.2.1/32 is directly connected, Serial0/0/0
C       172.16.3.0/24 is directly connected, GigabitEthernet0/0
L       172.16.3.1/32 is directly connected, GigabitEthernet0/0
L       172.16.3.1/32 is directly connected, GigabitEthernet0/0
S       192.168.1.0/24 [1/0] via 172.16.2.2
S       192.168.2.0/24 [1/0] via 172.16.2.2
R1#
```

```
R1# show cdp neighbors
```

Capability Codes: R - Router, T - Trans Bridge,
B - Source Route Bridge, S - Switch, H - Host,
I - IGMP, r - Repeater, P - Phone, D - Remote,
C - CVTA, M - Two-port Mac Relay

Device ID	Local Intrfce	Holdtime	Capability	Platform	Port ID
netlab-cs5	Gig 0/0	156	S I	WS-C2960-	Fas 0/1
R2	Ser 0/0/0	153	R S I	CIISCO1941	Ser 0/0/0

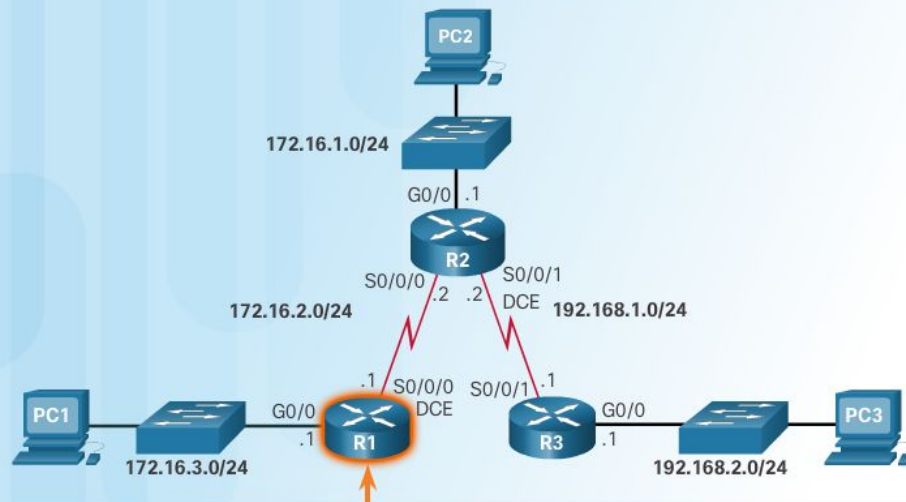
Troubleshoot IPv4 Static and Default Route Configuration

Solve a Connectivity Problem



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Verify Connectivity to the R3 LAN



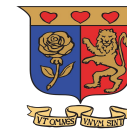
```
R1# ping 192.168.2.1 source g0/0
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.2.1, timeout is 2 seconds:
Packet sent with a source address of 172.16.3.1
.....
Success rate is 0 percent (0/5)
R1#
```

```
R1# traceroute 192.168.2.1
Type escape sequence to abort.
Tracing the route to 192.168.2.1
VRF info: (vrf in name/id, vrf out name/id)
 1 172.16.2.2 4 msec 4 msec 8 msec
 2 172.16.2.1 12 msec 12 msec 12 msec
 3 172.16.2.2 12 msec 8 msec 8 msec
 4 172.16.2.1 20 msec 16 msec 20 msec
 5 172.16.2.2 16 msec 16 msec 16 msec
 6 172.16.2.1 20 msec 20 msec 24 msec
 7 172.16.2.2 20 msec
```

A traceroute reveals that R2 forwards the traceroute back to R1. R1 returns it to R2

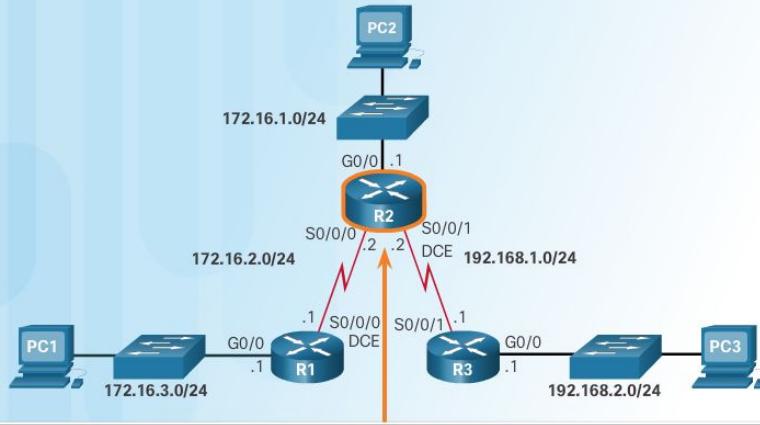
Troubleshoot IPv4 Static and Default Route Configuration

Solve a Connectivity Problem (Cont.)



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Verify the Routing Table



```
172.16.0.0/16 is variably subnetted, 5 subnets, 2 masks
C    172.16.1.0/24 is directly connected, GigabitEthernet0/0
L    172.16.1.1/32 is directly connected, GigabitEthernet0/0
C    172.16.2.0/24 is directly connected, Serial0/0/0
L    172.16.2.2/32 is directly connected, Serial0/0/0
S    172.16.3.0/24 is directly connected, Serial0/0/0
192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.1.0/24 is directly connected, Serial0/0/1
L    192.168.1.2/32 is directly connected, Serial0/0/1
S    192.168.2.0/24 [1/0] via 172.16.2.1
R2#
```

Static route to 192.168.2.0/24 has been configured using the next-hop address 172.16.2.1.

```
R2# show running-config | section ip route
ip route 172.16.3.0 255.255.255.0 172.16.2.1
ip route 192.168.2.0 255.255.255.0 172.16.2.1
R2#
R2# conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)# no ip route 192.168.2.0 255.255.255.0 172.16.2.1
R2(config)# ip route 192.168.2.0 255.255.255.0 192.168.1.1
R2(config)#
```

Incorrect route is removed and the correct route is then entered

```
R1# ping 192.168.2.1 source g0/0
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.2.1, timeout is 2 seconds:
Packet sent with a source address of 172.16.3.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 28/28/28 ms
R1#
```



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Dynamic Routing Protocols

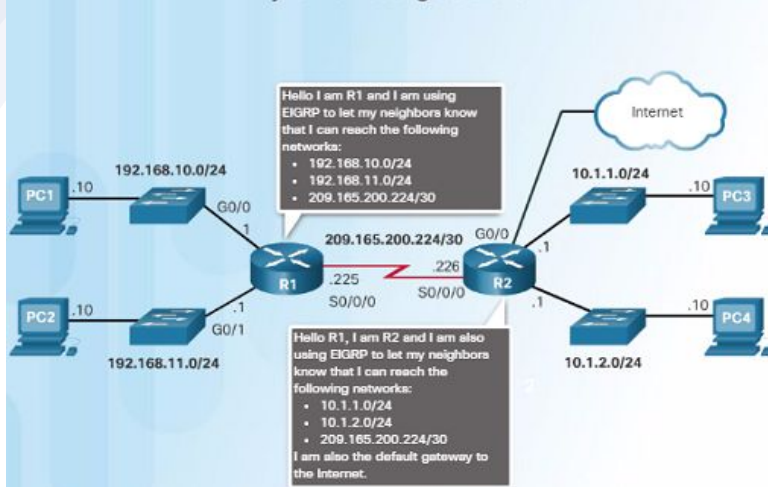
Dynamic Routing Protocols: Dynamic Routing



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- Dynamic routing protocols are used by routers to share information about the reachability and status of remote networks.
- Rather than manually configuring static routes, dynamic routing protocols use network discovery to share information about the networks that it knows about with other routers that are using the same routing protocol.
 - Routers automatically learn about remote networks from other routers
 - These networks and the best path to each are added to the routing table of the router.
- Routers have converged after they have finished exchanging and updating their routing tables. Routers then maintain the networks in their routing tables.

Dynamic Routing Scenario



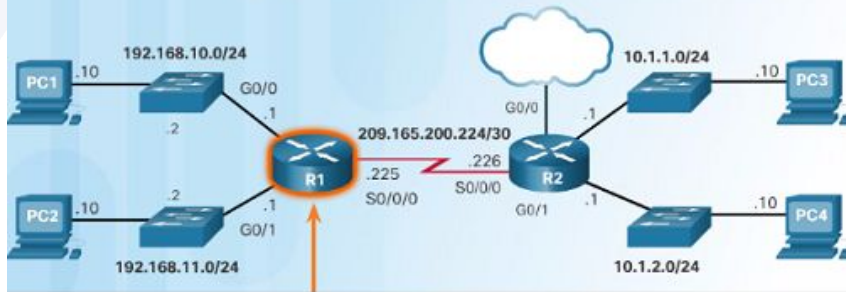
Dynamic Routing Protocols: IPv4 Routing Protocols



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- One of the major advantages of dynamic routing protocols over static routes -determine a new best path if the initial path becomes unusable.
- Dynamic routing protocols can adjust to topology changes without involving the network administrator.
- Cisco routers support a variety of IPv4 routing protocols including:
 - EIGRP
 - OSPF
 - IS-IS
 - RIP
 - Use **router ?** in global config mode to see the complete list.

Supported IPv4 Routing Protocols



```
R1(config)# router ?
bgp      Border Gateway Protocol (BGP)
eigrp    Enhanced Interior Gateway Routing Protocol (EIGRP)
isis     ISO IS-IS
iso-igrp  IGRP for OSI networks
mobile   Mobile routes
odr      On Demand stub Routes
ospf     Open Shortest Path First (OSPF)
ospfv3   OSPFv3
rip      Routing Information Protocol (RIP)
```

```
R1(config)# router
```

Dynamic Routing Protocol Overview

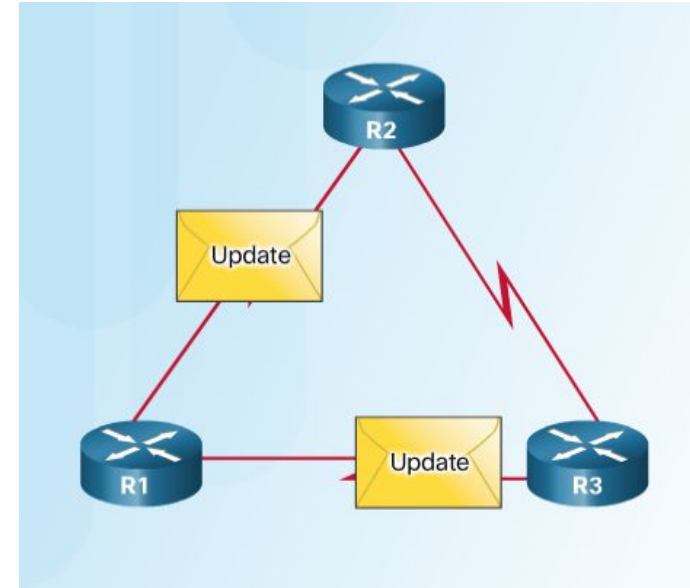
- RIP protocol was updated to RIPv2 to accommodate growth in the network environment
 - RIPv2 does not scale to current larger network implementations
- Routing Protocols developed to meet the need of larger networks include:
 - Open Shortest Path First (OSPF)
 - Intermediate System-to-Intermediate System (IS-IS).
 - Enhanced IGRP (EIGRP)
- Border Gateway Protocol (BGP) is used between Internet service providers (ISPs)

	Interior Gateway Protocols				Exterior Gateway Protocols
	Distance Vector		Link-State		Path Vector
IPv4	RIPv2	EIGRP	OSPFv2	IS-IS	BGP-4
IPv6	RIPng	EIGRP for IPv6	OSPFv3	IS-IS for IPv6	BGP-MP

Dynamic Routing Protocol Components



- Purpose of dynamic routing protocols includes:
 - Discovery of remote networks
 - Maintaining up-to-date routing information
 - Choosing the best path to destination networks
 - Ability to find a new best path if the current path is no longer available
- The main components of dynamic routing protocols include:
 - Data structures - tables or databases kept in RAM.
 - Routing protocol messages - to discover neighboring routers, exchange routing information, and maintain accurate information about the network.
 - Algorithms – to facilitate learning routing information and for best path determination.



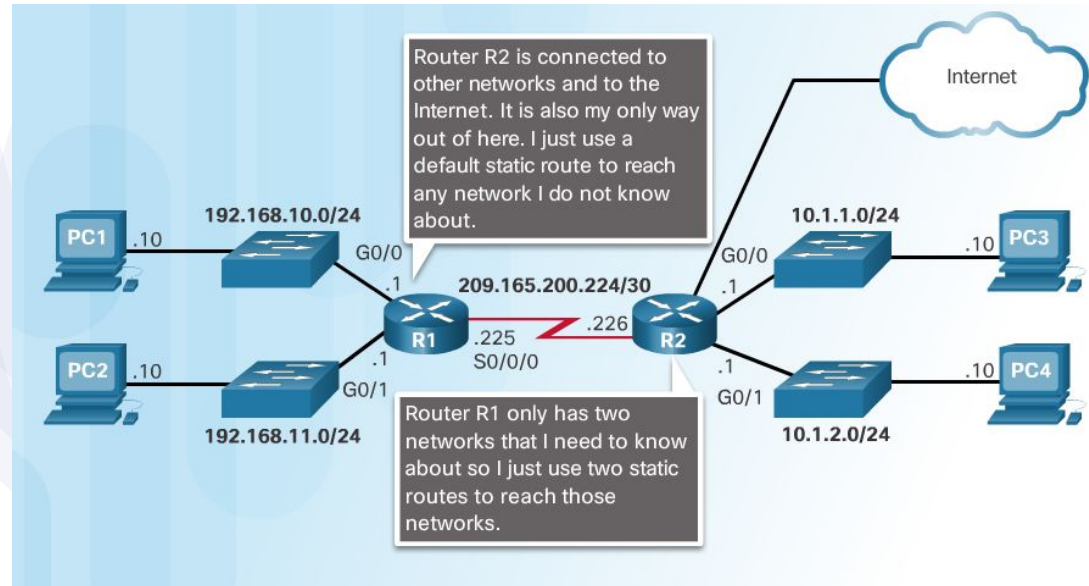
Dynamic versus Static Routing

Static Routing Uses



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- Networks often use both static and dynamic routing.
- Static Routing is used as follows:
 - For easy routing table maintenance in small networks.
 - Routing to and from a stub network.
 - Accessing a single default route.





Advantages	Disadvantages
Easy to implement in a small network.	Suitable only for simple topologies or for special purposes such as a default static route.
Very secure. No advertisements are sent as compared to dynamic routing protocols.	Configuration complexity increases dramatically as network grows.
Route to destination is always the same.	Manual intervention required to re-route traffic.
No routing algorithm or update mechanism required; therefore, extra resources (CPU or RAM) are not required.	

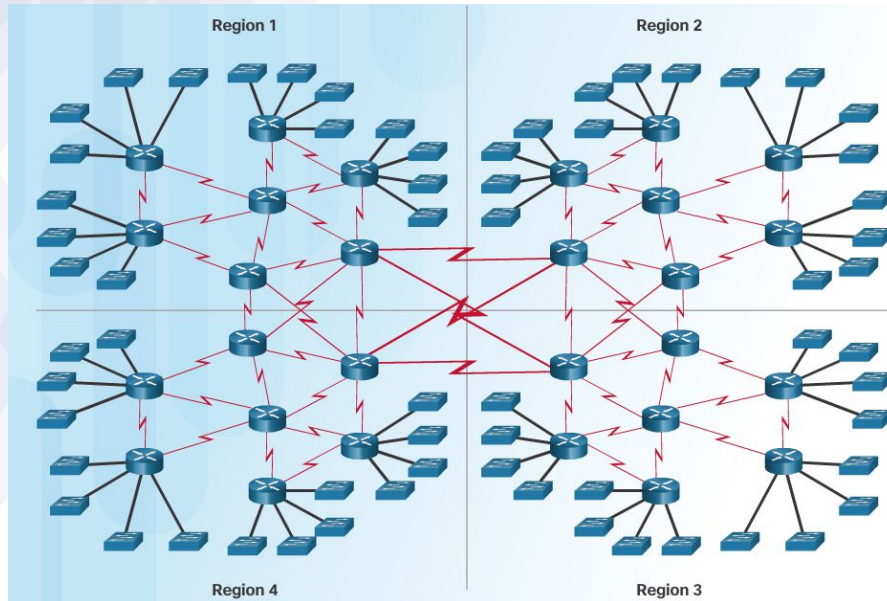
Dynamic versus Static Routing

Dynamic Routing Protocols Uses



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- Dynamic routing is the best choice for large networks
- Dynamic routing protocols help the network administrator manage the network:
 - Providing redundant paths
 - Automatically implementing the alternate path when a link goes down.



Dynamic Routing Advantages and Disadvantages



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Advantages

Suitable in all topologies where multiple routers are required.

Generally independent of the network size.

Automatically adapts topology to reroute traffic if possible.

Disadvantages

Can be more complex to implement.

Less secure. Additional configuration settings are required to secure.

Route depends on the current topology.

Requires additional CPU, RAM, and link bandwidth.



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RIPv2

Routing Information Protocol (RIP)



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- IP RIP (Routing Information Protocol) comes in two different versions: 1 and 2. Version 1 is a distance vector protocol (RFC 1058) and Version 2 is a hybrid protocol (RFCs 1721 and 1722).
- **Routing Information Protocol Version 1:** RIPv1 uses local broadcasts to share routing information. These updates are periodic in nature, occurring, by default, every 30 seconds. To prevent packets from circling around a loop forever, both versions of RIP solve counting to infinity by placing a hop count limit of 15 hops on packets. Any packet that reaches the sixteenth hop will be dropped. RIPv1 is a classful protocol. RIP supports up to six equal-cost paths to a single destination. Equal-cost path are the paths where the metric is same (Hop count).
- **Routing Information Protocol Version 2:** RIPv2 is a distance vector routing protocol with routing enhancements built into it, and it is based on RIPv1. Therefore, it is commonly called as hybrid routing protocol.

Routing Information Protocol (RIP)



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- RIPv2 uses multicasts instead of broadcasts. RIPv2 supports triggered updates. when a change occurs, a RIPv2 router will immediately propagate its routing information to its connected neighbours.
- RIPv2 is a classless protocol and it supports variable-length subnet masking (VLSM). Both RIPv1 and RIPv2 uses hop count as the metric.

Differences between RIPv1 and RIPv2

- RIPv1
 - Supports only classful routing (Does not support VLSM).
 - No authentication.
 - RIPv1 uses Broadcast.

Routing Information Protocol (RIP)



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- RIPv2
 - Supports classless routing (Supports VLSM and CIDR). Incorporates the addition of the network mask in the update to allow classless routing advertisements.
 - Authentication is available.
 - RIPv2 uses multi-cast instead of broadcast. multicast communication reduces the burden on the network devices that do not need to listen to

```
Router>enable
```

```
Router#configure terminal
```

```
Router(config)# router rip
```

```
Router(config-router)# version 2
```

```
Router(config-router)# network network_id
```

```
Router02>enable
Router02#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router02 (config)#router rip
Router02 (config-router)#version 2
Router02 (config-router)#network 172.17.0.0
Router02 (config-router)#network 172.18.0.0
Router02 (config-router)#network 172.19.0.0
Router02 (config-router)#exit
Router02 (config)#exit
Router02#
```

Dynamic versus Static Routing

Router RIP Configuration Mode



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- Use the **router rip** command to enable RIP
- Use the **no router rip** command to disable

```
R1# conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)# router rip
R1(config-router)#
```

RIP Configuration Options

```
R1(config-router)# ?

Router configuration commands:
address-family      Enter Address Family command mode
auto-summary        Enable automatic network number summarization
default             Set a command to its defaults
default-information  Control distribution of default information
default-metric      Set metric of redistributed routes
distance            Define an administrative distance
distribute-list      Filter networks in routing updates
exit                Exit from routing protocol configuration mode
flash-update-threshold Specify flash update threshold in second
help                Description of the interactive help system
input-queue         Specify input queue depth
maximum-paths        Forward packets over multiple paths
neighbor            Specify a neighbor router
network             Enable routing on an IP network
no                  Negate a command or set its defaults
offset-list          Add or subtract offset from RIP metrics
output-delay         Interpacket delay for RIP updates
passive-interface    Suppress routing updates on an interface
redistribute         Redistribute information from another routing protocol
timers              Adjust routing timers
traffic-share         How to compute traffic share over alternate paths
validate-update-source Perform sanity checks against source address of routing updates
version             Set routing protocol version

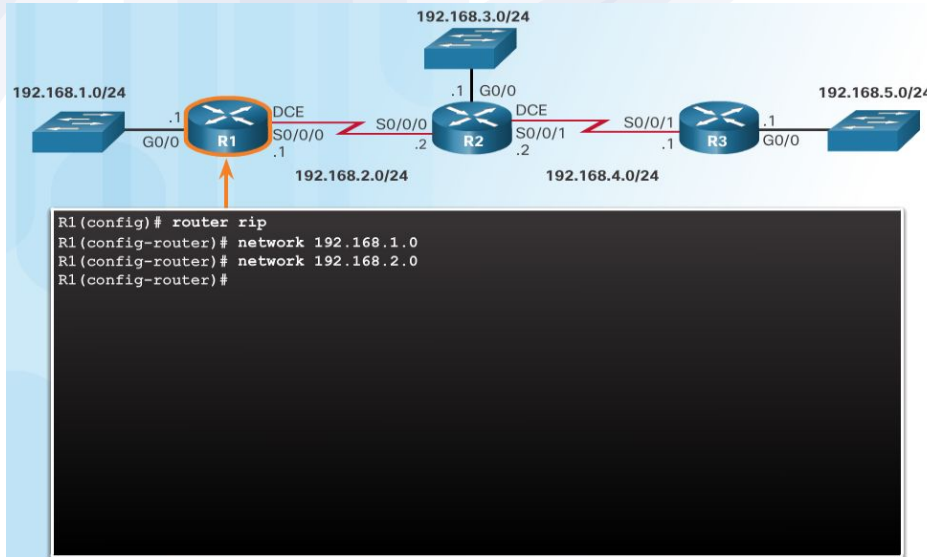
R1(config-router)#
```

Configuring the RIP Protocol Advertise Networks



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- The **network** *network-address* router configuration mode command:
 - Enables RIP on all interfaces that belong to a specific network
 - Advertises the network in RIP routing updates sent to other routers every 30 seconds.



Note: RIPv1 is a classful routing protocol for IPv4.

Configuring the RIP Protocol

Verify RIP Routing



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```
R1# show ip protocols
*** IP Routing is NSF aware ***

Routing Protocol is "rip"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Sending updates every 30 seconds, next due in 16 seconds
  Invalid after 180 seconds, hold down 180, flushed after 240
  Redistributing: rip

  Default version control: send version 1, receive any version
    Interface        Send Recv Triggered RIP Key-chain
  GigabitEthernet0/0  1     1  2
  Serial0/0/0        1     1  2

Automatic network summarization is in effect:
Maximum path: 4
Routing for Networks:
  192.168.1.0
  192.168.2.0

Routing Information Sources:
  Gateway         Distance    Last Update
  192.168.2.2      120        00:00:15
Distance: (default is 120)
```

R1#

show ip protocols – displays IPv4 routing protocols configured on the router.

```
R1# show ip route | begin Gateway
Gateway of last resort is not set

    192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
  C    192.168.1.0/24 is directly connected,
  GigabitEthernet0/0
  L    192.168.1.1/32 is directly connected,
  GigabitEthernet0/0
    192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
  C    192.168.2.0/24 is directly connected, Serial0/0/0
  L    192.168.2.1/32 is directly connected, Serial0/0/0
  R    192.168.3.0/24 [120/1] via 192.168.2.2, 00:00:24,
  Serial0/0/0
  R    192.168.4.0/24 [120/1] via 192.168.2.2, 00:00:24,
  Serial0/0/0
  R    192.168.5.0/24 [120/2] via 192.168.2.2, 00:00:24,
  Serial0/0/0
R1#
```

show ip route – displays RIP routes installed in the routing table.

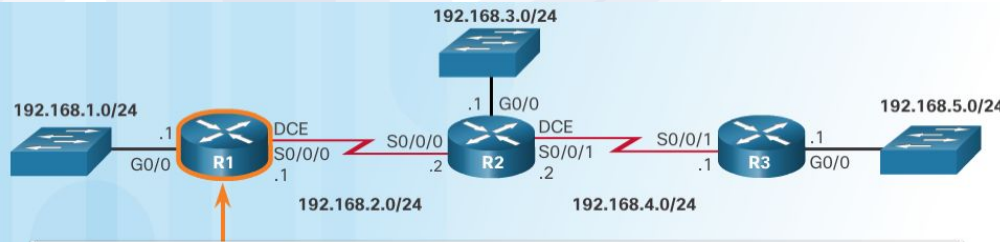
Configuring the RIP Protocol

Enable and Verify RIPv2



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- Use the **version 2** router configuration mode command to enable RIPv2
- Use the **show ip protocols** command to verify that RIPv2 is configured.
- Use the **show ip route** command to verify the RIPv2 routes in the routing table.



```
R1(config)# router rip
R1(config-router)# version 2
R1(config-router)# ^Z
R1#
R1# show ip protocols | section Default
Default version control: send version 2, receive version 2
Interface      Send  Recv  Triggered RIP  Key-chain
GigabitEthernet0/0    2    2
Serial0/0/0          2    2
R1#
```

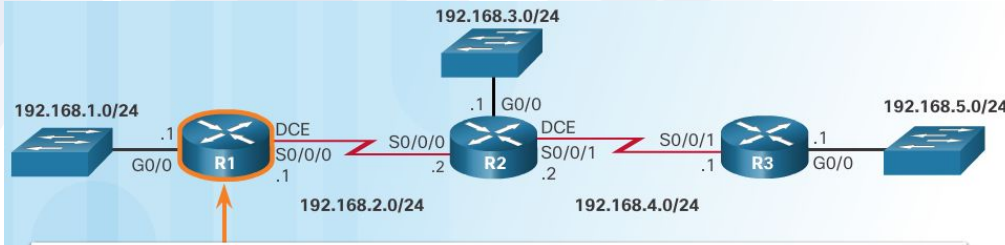
Configuring the RIP Protocol

Disable Auto Summarization



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- RIPv2 automatically summarizes networks at major network boundaries.
- Use the **no auto-summary** router configuration mode command to disable auto summarization.
- Use the **show ip protocols** command to verify that auto summarization is off.



```
R1(config)# router rip
R1(config-router)# no auto-summary
R1(config-router)# end
R1#
*Mar 10 14:11:49.659: %SYS-5-CONFIG_I: Configured from
console by console
R1# show ip protocols | section Automatic
Automatic network summarization is not in effect
R1#
```

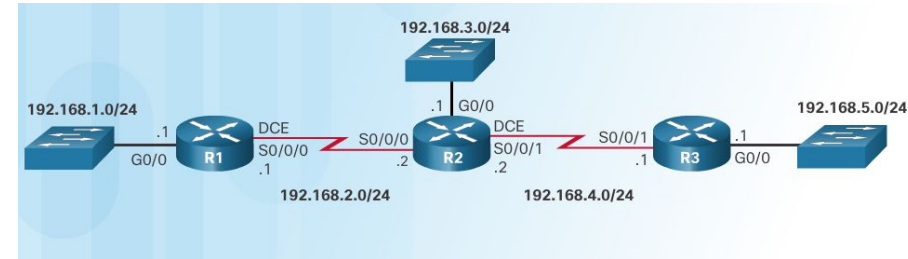
Configuring the RIP Protocol

Configure Passive Interfaces



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- RIP updates:
 - Are forwarded out all RIP-enabled interfaces by default.
 - Only need to be sent out interfaces that are connected to other RIP-enabled routers.
- Sending RIP updates to LANs wastes bandwidth, wastes resources, and is a security risk.
- Use the **passive-interface** router configuration command to stop routing updates out the interface. Still allows that network to be advertised to other routers.



```
R1(config)# router rip
R1(config-router)# passive-interface g0/0
R1(config-router)# end
R1#
R1# show ip protocols | begin Default
  Default version control: send version 2, receive version 2
  Interface                Send Recv Triggered RIP Key-
chain
  Serial0/0/0                2    2
Automatic network summarization is not in effect
Maximum path: 4
Routing for Networks:
  192.168.1.0
  192.168.2.0
Passive Interface(s):
  GigabitEthernet0/0
Routing Information Sources:
  Gateway      Distance    Last Update
  192.168.2.2    120        00:00:06
Distance: (default is 120)

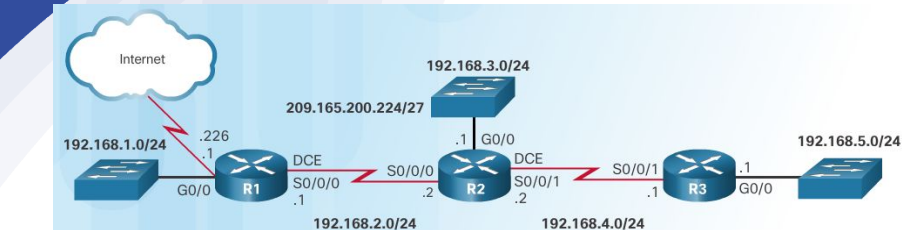
R1#
```

Configuring the RIP Protocol Propagate a Default Route



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- In the diagram a default static route to the Internet is configured on R1.
- The **default-information originate** router configuration command instructs R1 to send the default static route information in the RIP updates.



```
R1(config)# ip route 0.0.0.0 0.0.0.0 S0/0/1 209.165.200.226
R1(config)# router rip
R1(config-router)# default-information originate
R1(config-router)# ^Z
R1#
*Mar 10 23:33:51.801: %SYS-5-CONFIG_I: Configured from console by console
R1# show ip route | begin Gateway
Gateway of last resort is 209.165.200.226 to network 0.0.0.0

S* 0.0.0.0/0 [1/0] via 209.165.200.226, Serial0/0/1
    192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.1.0/24 is directly connected, GigabitEthernet0/0
L    192.168.1.1/32 is directly connected, GigabitEthernet0/0
    192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.2.0/24 is directly connected, Serial0/0/0
L    192.168.2.1/32 is directly connected, Serial0/0/0
R    192.168.3.0/24 [120/1] via 192.168.2.2, 00:00:08, Serial0/0/0
R    192.168.4.0/24 [120/1] via 192.168.2.2, 00:00:08, Serial0/0/0
R    192.168.5.0/24 [120/2] via 192.168.2.2, 00:00:08, Serial0/0/0
    209.165.200.0/24 is variably subnetted, 2 subnets, 2 masks
C    209.165.200.0/24 is directly connected, Serial0/0/1
L    209.165.200.225/27 is directly connected, Serial0/0/1
R1#
```

Dynamic Routing Protocols: RIPng



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- The primary features of RIPng are the same as they were in IPv4: It's still a *distance-vector* protocol, has a **maximum hop** count of **15** and uses **split horizon**, **poison reverse** and other loop avoidance mechanisms.
- RIPng still uses **multicast** to send routing updates, but in IPv6, it uses **FF02::9** for the multicast traffic.
- But there are also some differences between the two versions. One of the most notable changes with RIPng (and all other IPv6 routing protocols), is that you enable network advertisement from interface configuration mode of a router instead of using **network** command in the global configuration mode.

Dynamic Routing Protocols: RIPng



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Router(config-if)#ipv6 rip 10 enable

- The **10** that you see in the command is a number that identifies the **process** of RIPng that's running. Just by typing this command, a new RIPng process is started (enabled) for you so you don't have to go to global configuration mode to enable the process.

```
R1(config)#int fa0/0
```

```
R1(config-if)#ipv6 RIP 10 enable
```

```
R1(config-if)#
```

```
R1(config-if)#int serial 0/0/0
```

```
R1(config-if)#ipv6 RIP 10 enable
```

SUMMARY STEPS

- enable**
- configure terminal**
- ipv6 unicast-routing**
- interface type number**
- ipv6 rip name enable**

Step 1	enable Example: Router#enable	Enables privileged EXEC mode. • Enter your password if prompted.
Step 2	configure terminal Example: Router#configure terminal	Enters global configuration mode.
Step 3	ipv6 router rip word Example: Router(config)#ipv6 router rip process1	Configures an IPv6 RIP routing process and enters router configuration mode for the IPv6 RIP routing process. • Use the wordargument to identify a specific IPv6 RIP routing process.
Step 4	maximum-paths number-paths Example: Router(config-router)#maximum-paths 1	(Optional) Defines the maximum number of equal-cost routes that IPv6 RIP can support. • The number-paths argument is an integer from 1 to 64. The default for RIP is four paths.
Step 5	exit Example: Router(config-if)#exit	Exits interface configuration mode and enters global configuration mode.
Step 6	interface type number Example: Router(config)#interface gigabitethernet 0/0/0	Specifies the interface type and number, and enters interface configuration mode.
Step 7	ipv6 rip name default-information (only) originate [metric metric-value] Example: Router(config-if)#ipv6 rip process1 default-information originate	(Optional) Originates the IPv6 default route (::/0) into the specified RIP routing process updates sent out of the specified interface. Note To avoid routing loops after the IPv6 default route (::/0) is originated out of any interface, the routing process ignores all default routes received on any interface. • Specifying the only keyword originates the default route (::/0) but suppresses all other routes in the updates sent on this interface. • Specifying the originate keyword originates the default route (::/0) in addition to all other routes in the updates sent on this interface.

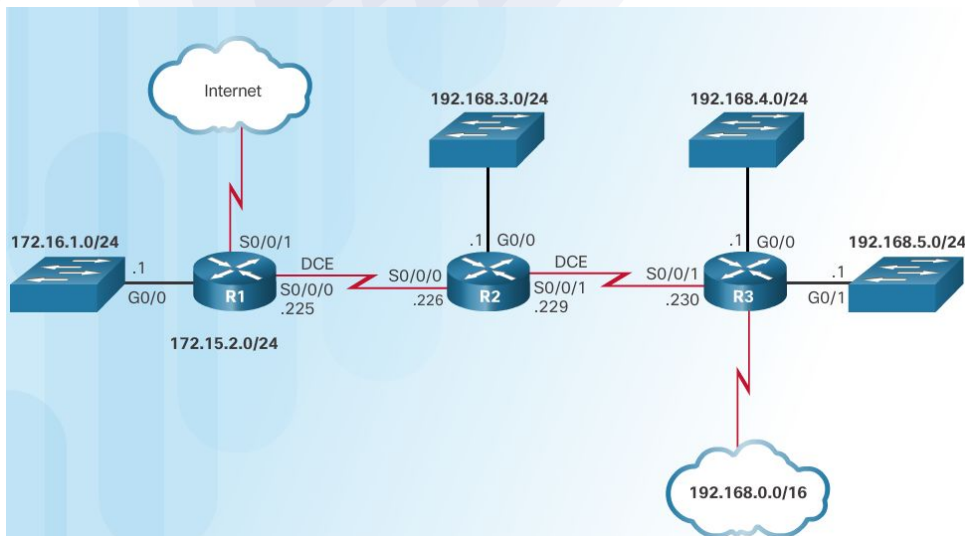


The Routing Table

Parts of an IPv4 Route Entry



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```
R1# show ip route | begin Gateway
Gateway of last resort is 209.165.200.234 to network 0.0.0.0

S* 0.0.0.0/0 [1/0] via 209.165.200.234, Serial0/0/1
    is directly connected, Serial0/0/1
C 172.16.0.0/16 is variably subnetted, 5 subnets, 3 masks
L 172.16.1.0/24 is directly connected, GigabitEthernet0/0
L 172.16.1.1/32 is directly connected, GigabitEthernet0/0
R 172.16.2.0/24 [120/1] via 209.165.200.226, 00:00:12, Serial0/0/0
R 172.16.3.0/24 [120/2] via 209.165.200.226, 00:00:12, Serial0/0/0
R 172.16.4.0/28 [120/2] via 209.165.200.226, 00:00:12, Serial0/0/0
R 192.168.0.0/16 [120/2] via 209.165.200.226, 00:00:03, Serial0/0/0
    209.165.200.0/24 is variably subnetted, 5 subnets, 2 masks
C 209.165.200.224/30 is directly connected, Serial0/0/0
L 209.165.200.225/32 is directly connected, Serial0/0/0
R 209.165.200.228/30 [120/1] via 209.165.200.226, 00:00:12, Serial0/0/0
C 209.165.200.232/30 is directly connected, Serial0/0/1
L 209.165.200.233/30 is directly connected, Serial0/0/1
R1#
```

Routing Table for R1

Parts of an IPv4 Route Entry

Directly Connected Entries



- Directly Connected Networks (C) are automatically added to the routing table when the interface is configured and activated.
- Entries contain the following information:
 - Route source - how the route was learned.
 - Destination network – remote network.
 - Outgoing Interface – exit interface used to forward packets to destination.
- Other route source entries include:
 - S –Static Route
 - D – EIGRP routing protocol
 - O – OSPF routing protocol
 - R - RIP routing protocol

Route Source	Destination Network	Outgoing Interface
C	172.16.1.0/24 is directly connected,	GigabitEthernet0/0
L	172.16.1.1/32 is directly connected,	GigabitEthernet0/0

Legend

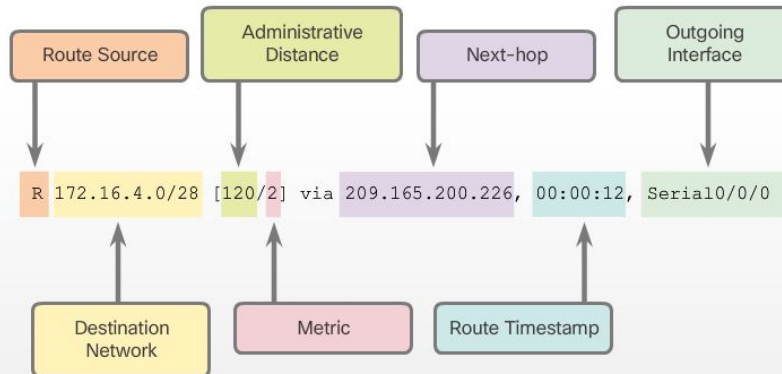
-  - Identifies how the network was learned by the router.
-  - Identifies the destination network and how it is connected.
-  - Identifies the interface on the router connected to the destination network.

Parts of an IPv4 Route Entry

Remote Network Entries



- Routes to remote networks contain the following information:
 - Route source – how route was learned
 - Destination network
 - Administrative distance (AD) - trustworthiness of the route.
 - Metric – value assigned to reach the remote network. Lower is better.
 - Next hop – IPv4 address of the next router that the packet should be forwarded to.
 - Route timestamp – time since the route was updated.
 - Outgoing interface - the exit interface to use to forward the packet



Dynamically Learned IPv4 Routes

Routing Table Terms



- The routing table is a hierarchical structure that is used to speed up the lookup process when locating routes and forwarding packets.
- The hierarchy includes:
 - Ultimate Routes
 - Level 1 routes
 - Level 1 parent routes
 - Level 2 child routes

```
R1# show ip route | begin Gateway
Gateway of last resort is 209.165.200.234 to network 0.0.0.0

S*    0.0.0.0/0 [1/0] via 209.165.200.234, Serial0/0/1
      172.16.0.0/16 is variably subnetted, 5 subnets, 3 masks
C     172.16.1.0/24 is directly connected, GigabitEthernet0/0
L     172.16.1.1/32 is directly connected, GigabitEthernet0/0
R     172.16.2.0/24 [120/1] via 209.165.200.226, 00:00:12, Serial0/0/0
R     172.16.3.0/24 [120/2] via 209.165.200.226, 00:00:12, Serial0/0/0
R     172.16.4.0/28 [120/2] via 209.165.200.226, 00:00:12, Serial0/0/0
R     192.168.0.0/16 [120/2] via 209.165.200.226, 00:00:03, Serial0/0/0
      209.165.200.0/24 is variably subnetted, 5 subnets, 2 masks
C     209.165.200.224/30 is directly connected, Serial0/0/0
L     209.165.200.225/32 is directly connected, Serial0/0/0
R     209.165.200.228/30 [120/1] via 209.165.200.226, 00:00:12, Serial0/0/0
C     209.165.200.232/30 is directly connected, Serial0/0/1
L     209.165.200.233/32 is directly connected, Serial0/0/1
R1#
```

Dynamically Learned IPv4 Routes

Ultimate Route



- An ultimate route is a routing table entry that contains either a next-hop IPv4 address or an exit interface.
- Directly connected, dynamically learned, and local routes are ultimate routes.

```
R1# show ip route | begin Gateway
Gateway of last resort is 209.165.200.234 to network 0.0.0.0

S*   0.0.0.0/0 [1/0] via 209.165.200.234, Serial0/0/1
      is directly connected, Serial0/0/1
      172.16.0.0/16 is variably subnetted, 5 subnets, 3 masks
      C   172.16.1.0/24 is directly connected, GigabitEthernet0/0
      L   172.16.1.1/32 is directly connected, GigabitEthernet0/0
      R   172.16.2.0/24 [120/1] via 209.165.200.226, 00:00:12, Serial0/0/0
      R   172.16.3.0/24 [120/2] via 209.165.200.226, 00:00:12, Serial0/0/0
      R   172.16.4.0/28 [120/2] via 209.165.200.226, 00:00:12, Serial0/0/0
      R   192.168.0.0/16 [120/2] via 209.165.200.226, 00:00:03, Serial0/0/0
      209.165.200.0/24 is variably subnetted, 5 subnets, 2 masks
      C   209.165.200.224/30 is directly connected, Serial0/0/0
      L   209.165.200.225/32 is directly connected, Serial0/0/0
      R   209.165.200.228/30 [120/1] via 209.165.200.226, 00:00:12, Serial0/0/0
      C   209.165.200.232/30 is directly connected, Serial0/0/1
      L   209.165.200.233/32 is directly connected, Serial0/0/1
R1#
```

Dynamically Learned IPv4 Routes

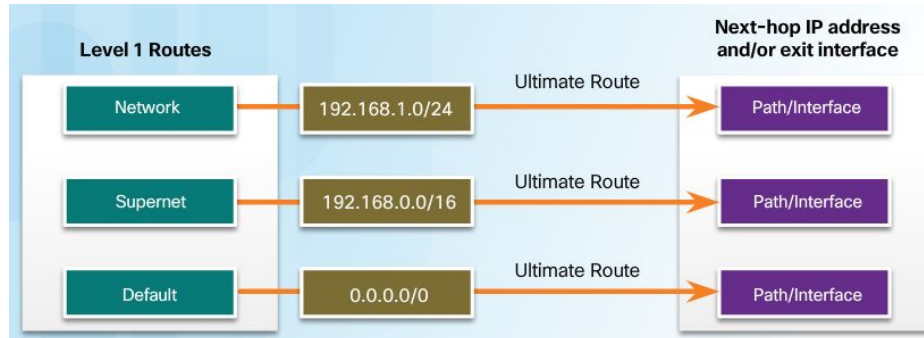
Level 1 Route



- A level 1 route can be a:
 - **Network route** - a network route that has a subnet mask equal to that of the classful mask.
 - **Supernet route** - a network address with a mask less than the classful mask, for example, a summary address.
 - **Default route** - a static route with the address 0.0.0.0/0

```
R1# show ip route | begin Gateway
Gateway of last resort is 209.165.200.234 to network 0.0.0.0

S*   0.0.0.0/0 [1/0] via 209.165.200.234, Serial0/0/1
      is directly connected, Serial0/0/1
C    172.16.0.0/16 is variably subnetted, 5 subnets, 3 masks
L    172.16.1.0/24 is directly connected, GigabitEthernet0/0
L    172.16.1.1/32 is directly connected, GigabitEthernet0/0
R    172.16.2.0/24 [120/1] via 209.165.200.226, 00:00:12, Serial0/0/0
R    172.16.3.0/24 [120/2] via 209.165.200.226, 00:00:12, Serial0/0/0
R    172.16.4.0/28 [120/2] via 209.165.200.226, 00:00:12, Serial0/0/0
R    192.168.0.0/16 [120/2] via 209.165.200.226, 00:00:03, Serial0/0/0
C    209.165.200.0/24 is variably subnetted, 5 subnets, 2 masks
C    209.165.200.224/30 is directly connected, Serial0/0/0
L    209.165.200.225/32 is directly connected, Serial0/0/0
R    209.165.200.228/30 [120/1] via 209.165.200.226, 00:00:12, Serial0/0/0
C    209.165.200.232/30 is directly connected, Serial0/0/1
L    209.165.200.233/32 is directly connected, Serial0/0/1
R1#
```



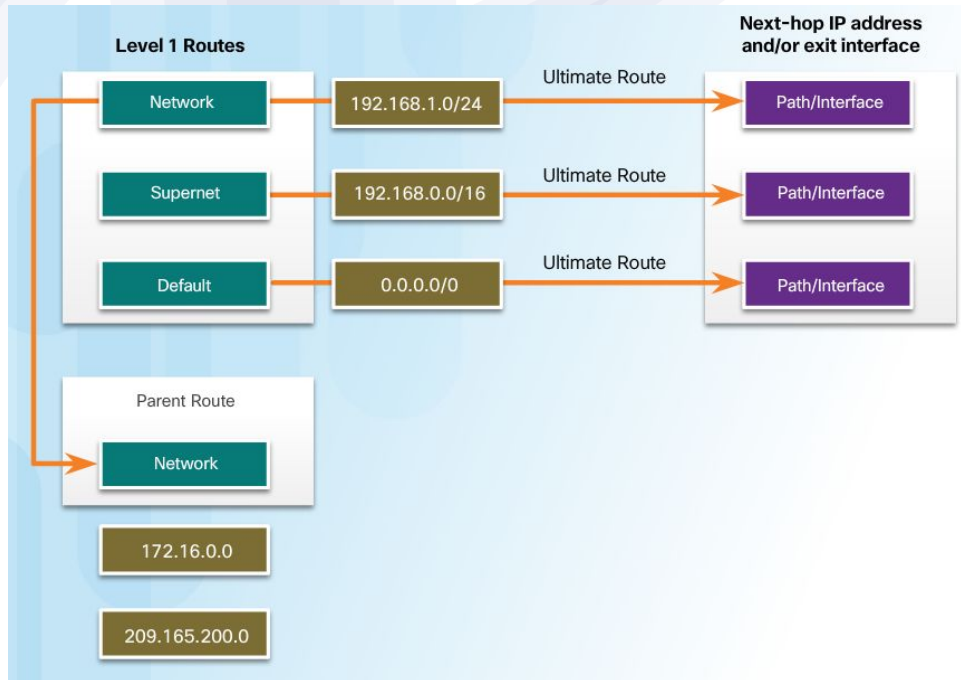
Dynamically Learned IPv4 Routes

Level 1 Parent Route



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- A parent route is a level 1 network route that is subnetted.
- In the routing table, it basically provides a heading for the specific subnets it contains.



```
R1# show ip route | begin Gateway
Gateway of last resort is 209.165.200.234 to network 0.0.0.0

S*   0.0.0.0/0 [1/0] via 209.165.200.234, Serial0/0/1
      is directly connected, Serial0/0/1
      172.16.0.0/16 is variably subnetted, 5 subnets, 3 masks
C     172.16.1.0/24 is directly connected, GigabitEthernet0/0
L     172.16.1.1/32 is directly connected, GigabitEthernet0/0
R     172.16.2.0/24 [120/1] via 209.165.200.226, 00:00:12, Serial0/0/0
R     172.16.3.0/24 [120/2] via 209.165.200.226, 00:00:12, Serial0/0/0
R     172.16.4.0/28 [120/2] via 209.165.200.226, 00:00:12, Serial0/0/0
R     192.168.0.0/16 [120/2] via 209.165.200.226, 00:00:03, Serial0/0/0
      209.165.200.0/24 is variably subnetted, 5 subnets, 2 masks
C     209.165.200.224/30 is directly connected, Serial0/0/0
L     209.165.200.225/32 is directly connected, Serial0/0/0
R     209.165.200.228/30 [120/1] via 209.165.200.226, 00:00:12, Serial0/0/0
C     209.165.200.232/30 is directly connected, Serial0/0/1
L     209.165.200.233/32 is directly connected, Serial0/0/1
R1#
```

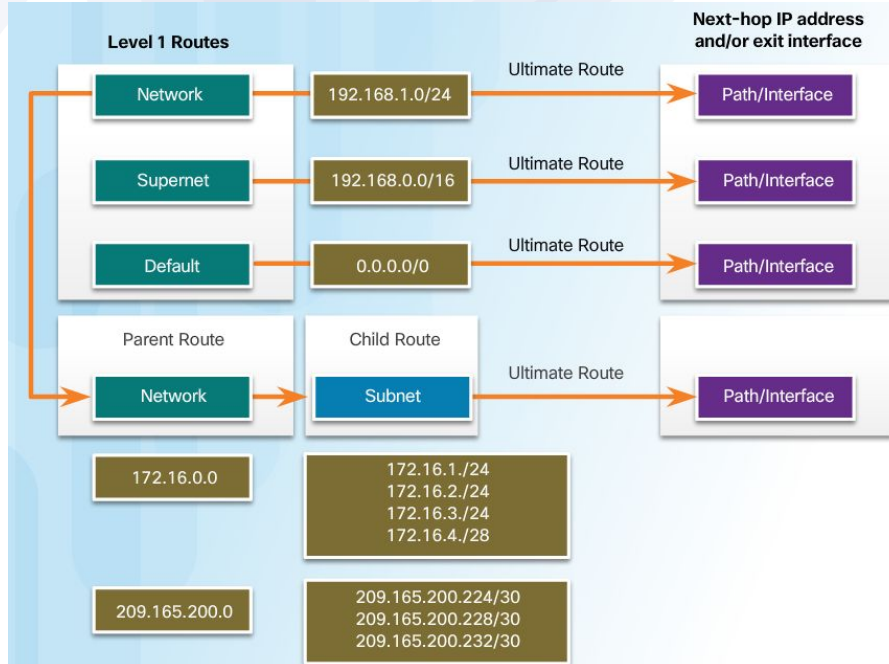
Dynamically Learned IPv4 Routes

Level 2 Child Route



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- A level 2 child route is a route that is a subnet of a classful network address.
- Level 1 parent routes contain level 2 child routes.
- Level 2 child routes are also ultimate routes.



```
R1# show ip route | begin Gateway
Gateway of last resort is 209.165.200.234 to network 0.0.0.0

S*   0.0.0.0/0 [1/0] via 209.165.200.234, Serial0/0/1
      is directly connected, Serial0/0/1
C    172.16.0.0/16 is variably subnetted, 5 subnets, 3 masks
L    172.16.1.0/24 is directly connected, GigabitEthernet0/0
L    172.16.1.1/32 is directly connected, GigabitEthernet0/0
R    172.16.2.0/24 [120/1] via 209.165.200.226, 00:00:12, Serial0/0/0
R    172.16.3.0/24 [120/2] via 209.165.200.226, 00:00:12, Serial0/0/0
R    172.16.4.0/28 [120/2] via 209.165.200.226, 00:00:12, Serial0/0/0
R    192.168.0.0/16 [120/2] via 209.165.200.226, 00:00:03, Serial0/0/0
C    209.165.200.0/24 is variably subnetted, 5 subnets, 2 masks
C    209.165.200.224/30 is directly connected, Serial0/0/0
L    209.165.200.225/32 is directly connected, Serial0/0/0
R    209.165.200.228/30 [120/1] via 209.165.200.226, 00:00:12, Serial0/0/0
C    209.165.200.232/30 is directly connected, Serial0/0/1
L    209.165.200.233/30 is directly connected, Serial0/0/1
R1#
```

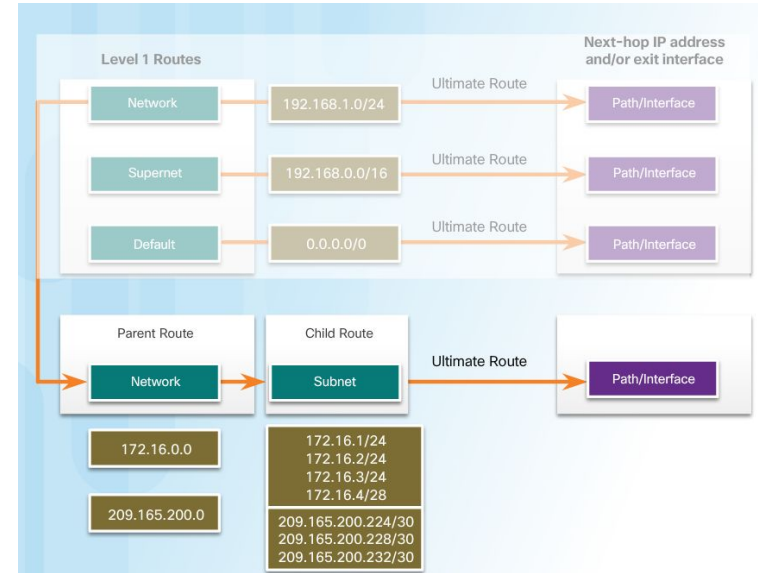
The IPv4 Route Lookup Process

Route Lookup Process



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- Router lookup process:
 - If the best match is a level 1 ultimate route, then this route is used to forward the packet.
 - If the best match is a level 1 parent route, the router then examines child routes (the subnet routes).
 - If there is a match with a level 2 child route, that is used to forward the packet.
 - If there is no match with level 2 child routes, the router searches level 1 supernet or default routes. If there is a match, that route is used.
 - If there is no match found in the routing table the packet is dropped.



The IPv4 Route Lookup Process

Best Route = Longest Match



- The best match is the route in the routing table that has the most number of far left matching bits with the destination IPv4 address of the packet.
- The route with the greatest number of equivalent far left bits, or the longest match, is always the preferred route.

IP Packet Destination	172.16.0.10	10101100.00010000.00000000.00001010
Route 1	172.16.0.0/12	10101100.00010000.00000000.00000000
Route 2	172.16.0.0/18	10101100.00010000.00000000.00000000
Route 3	172.16.0.0/26	10101100.00010000.00000000.00000000

Longest Match to IP Packet Destination

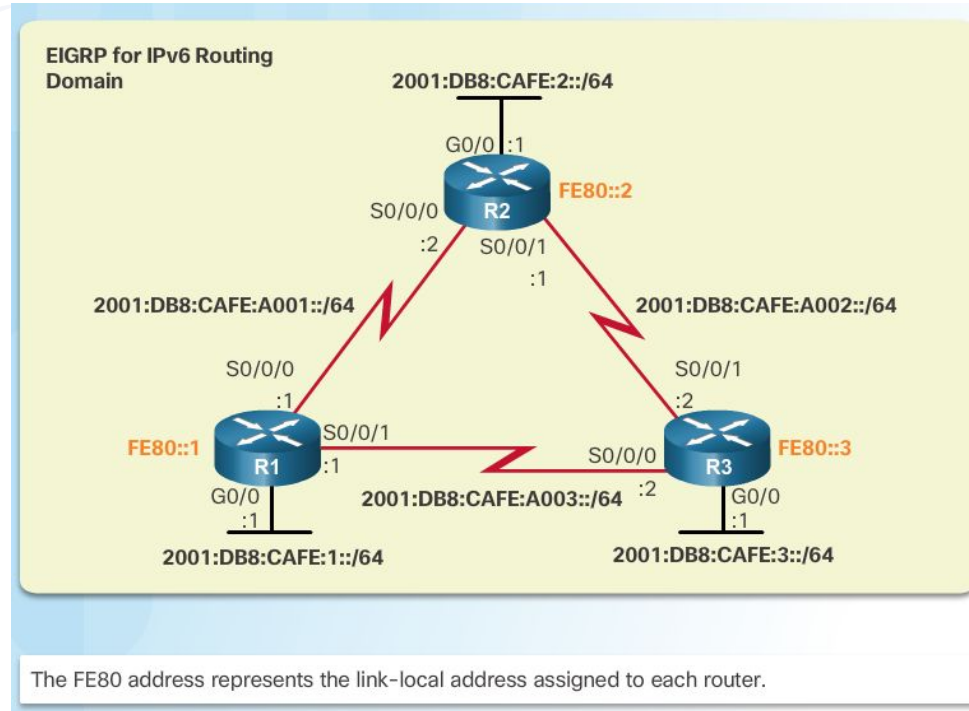
Analyze an IPv6 Routing Table

IPv6 Routing Table Entries



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- An IPv6 routing table includes directly connected, static and dynamically learned routes.
- All IPv6 routes are level 1 ultimate routes.



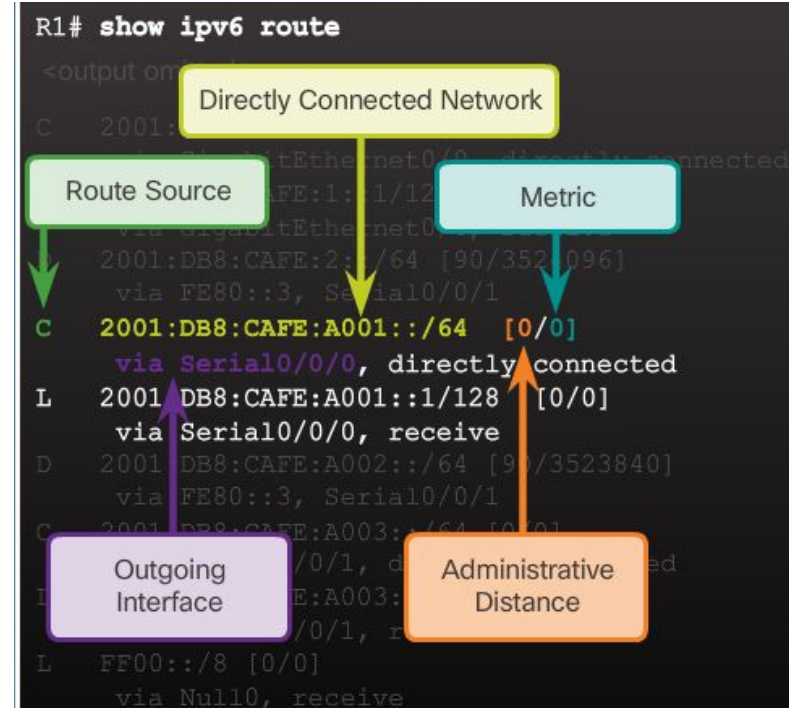
Analyze an IPv6 Routing Table

Directly Connected Entries



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- Use the **show ipv6 route** command to display the IPv6 routing table.
- The directly connected route entries include the following:
 - Route source – How the route was learned. Directly connected indicated with a C and L for local route.
 - Directly connected network address.
 - Administrative distance – Trustworthiness of the route (lower more trustworthy).
 - Metric – Value assigned to reach the network (lower is preferred route).
 - Outgoing interface – Exit interface used to forward packet.



Analyze an IPv6 Routing Table

Remote IPv6 Network Entries

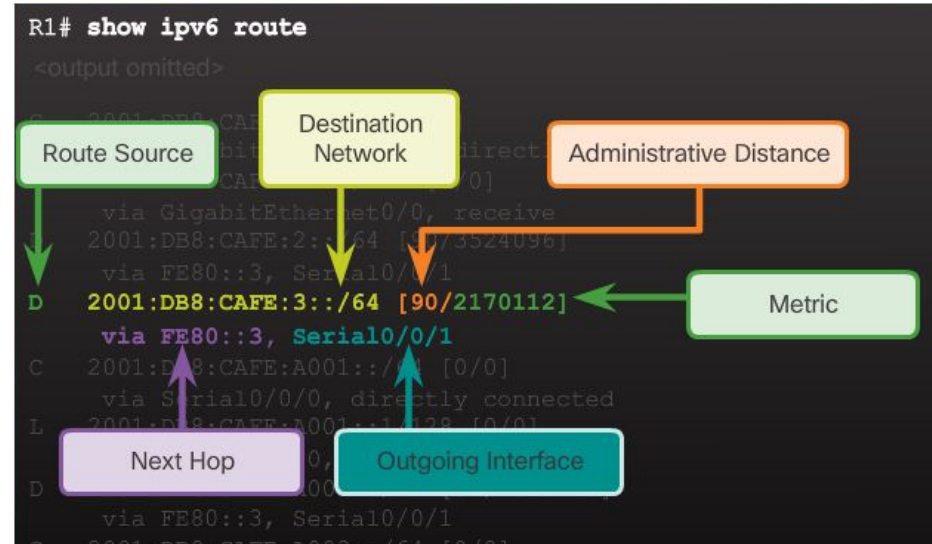


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- The remote IPv6 route entries also include the following:
 - Route source – How the route was learned. Common codes include O (OSPF), D (EIGRP), R (RIP), and S (Static route).
 - Next hop - Identifies the IPv6 address of the next router to forward the packet to.

The IPv6 router lookup process:

- Examines level 1 network routes for the best match.
- Longest match is the best match.





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